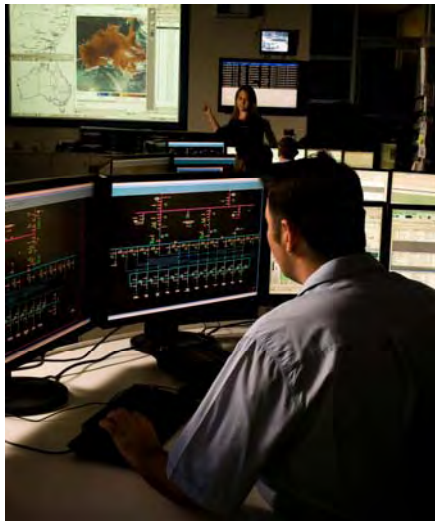




Electricity Network Performance Report 2008-2009

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Electricity Network Performance Report 2008-2009

Submitted to Industry and Investment NSW

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1. Profile

Country Energy is a leading Australian energy services provider owned by the New South Wales Government.

With annual revenue of more than \$2 billion, Country Energy is one of the nation's largest utility businesses. Our success is driven by commercial decisions that meet both local priorities and the needs of our customers, employees, stakeholders and shareholders.

Our core business is to operate a safe and reliable electricity, water and gas network service for our customers.

We operate Australia's largest electricity distribution network, extending across 95 per cent of New South Wales' land mass and into parts of Queensland, Victoria and the Australian Capital Territory. This network is unique in Australia, with medium to very low customer density and wide variations in topographical and climatic conditions. It includes:

- over 200,000 kilometres of powerlines, 97 per cent of which are overhead and 3 per cent underground
- around 1.4 million power poles and over 142,000 street lights
- some 130,000 distribution substations and over 300 zone substations. The majority of these substations are remotely monitored and controlled by Supervisory Control and Data Acquisition (SCADA) systems
- a diverse range of asset types, from single wire earth return (SWER) through to single and three phase, operating at voltage levels of 220kV, 132kV, 110kV, 66kV, 33kV, 22kV, 19.1kV, 12.7kV, 11kV, 6.6kV and 3.3kV
- delivery of around 12,000 GWh of energy per year.

Some network components have an age of greater than 70 years, 50 per cent of overhead powerlines are more than 40 years old, and the overall weighted average asset age is around 27 years.

We also provide reticulated natural gas to more than 25,000 customers in southern NSW, and water and sewerage services to 10,000 customers in far western New South Wales.

We provide competitive energy offers and value-added services – including renewable energy, LPG, energy efficiency advice, water management solutions and flexible payment options – to customers in six states and territories.

Country Energy continues to support a decentralised regional management structure to serve a customer base of around 800,000 customers. Our 4,200 employees live and work in local communities across rural and regional New South Wales, operating from 142 Customer and Field Service Centres, eight regional offices and several metropolitan locations to better respond to customer needs and expectations,

1. Profile



Figure 1: Country Energy's network distribution area.

2. Network Management

Table 1.1 - Network Operator Statistics

	Number at 30-6-08	Number at 30-6-09
Distribution Customer Numbers (Total)	780,222	786,241
Customer Numbers at Year End (North Western)	58,175	58,507
Customer Numbers at Year End (Southern ¹)	68,567	111,230
Customer Numbers at Year End (Central Western ¹)	73,095	91,205
Customer Numbers at Year End (South Western ¹)	74,780	-
Customer Numbers at Year End (Northern)	92,551	93,349
Customer Numbers at Year End (South Eastern ¹)	104,864	119,941
Customer Numbers at Year End (Far North Coast)	135,615	136,943
Customer Numbers at Year End (Mid North Coast)	152,940	154,274
Customer Numbers at Year End (Far West)	19,635	20,792
Maximum Demand (MW)	2,329	2,332
Energy Received by Dist Network to Year End GWh	12,817	13,076
Energy Distributed to Year End (Residential) GWh	4,762	4,839
Energy Distributed to Year End (Non-Residential including un-metered supplies) GWh	7,211	7,285
Energy Distributed to Year End (North Western) GWh	1,166	1,172
Energy Distributed to Year End (Southern ¹) GWh	1,256	1,857
Energy Distributed to Year End (Central Western ¹) GWh	2,103	2,632
Energy Distributed to Year End (South Western ¹) GWh	1,320	-
Energy Distributed to Year End (Northern) GWh	1,379	1,394
Energy Distributed to Year End (South Eastern ¹) GWh	1,126	1,396
Energy Distributed to Year End (Far North Coast) GWh	1,518	1,522
Energy Distributed to Year End (Mid North Coast) GWh	1,646	1,673
Energy Distributed to Year End (Far West) GWh	458	478
System Loss Factor (%)	7.0	7.9
Transmission System (km)	0	0
Transmission Substation (Number)	18	17
Sub Transmission System (km)	12,162	12,277
Substation - Zone (Number)	324	321
Substation - Distribution (Number)	131,482	132,212
High Voltage Overhead (km)	159,779	144,546 ²
High Voltage Underground (km)	1,901	1,795 ²
Low Voltage Overhead (km)	27,178	26,308 ^{2,3}
Low Voltage Underground (km)	4,113	4,897
Pole (Number)	1,371,818	1,373,649
Streetlights (Number)	134,931	143,753
Employees (Full Time Equivalent Number)	4,182	4,278
Contractors (Full Time Equivalent Number)	117	267

¹ Riverina and South Western Regions were combined in January 2009 to form the Southern Region with slight boundary changes to the South Eastern and Central Western Region.

² Differences in line length due to data correction made during the year.

³ Line length does not include LV services. Approx length of LV services is 14,000km.

2. Network Management

2.1 OVERVIEW

Country Energy manages a complex and geographically diverse power supply network. As previously outlined, our distribution area is one of the largest in the world. To ensure our decisions are localised and reflect the needs and concerns of the local population, the network is managed through a regional management structure.

Country Energy has put in place key management plans to meet our statutory and public responsibilities in providing a safe, reliable and sustainable electricity supply to our customers, including our:

- **Network Management Plan** – aims to establish a framework which ensures Country Energy's subtransmission and distribution system provides an adequate, reliable and safe supply of electricity of appropriate quality. The plan is revised regularly and both management and operations employees are audited to ensure work practices are in accordance with the objectives of the plan (See sections 3 and 4).
- **Public Electrical Safety Awareness Plan** – provides a framework and strategies to warn the public in Country Energy's area of operations of the hazards associated with electricity and, in particular, the hazards associated with overhead powerlines, and to provide them with simple but effective ways to minimise their risk exposure (See section 9).
- **Bush Fire Risk Management Plan** – aims to ensure that Country Energy's assets are managed in a way that will minimise the risk of bush fires, as well as protect our assets and maintain customer supply reliability at times of bush fire (See section 8).
- **Customer Installation Safety Plan** – ensures provision of safe electrical installations within Country Energy's network area and their safe connection to Country Energy's subtransmission and distribution system to help provide a safe working environment for our employees, service providers, contractors, customers and the general public (See section 6).

Country Energy's commitment to safety management procedures can also be seen in its continuous improvements in safety measures (See section 5).

2.2 AUDITS

During the 2008/2009 year KPMG carried out a number of audits under Country Energy's Internal Audit Plan.

The activities of the internal audit team are primarily based on Country Energy's risk profile and are directed to reviewing key processes and controls that have been established to mitigate key strategic risks.

Two classes of activity warrant further focus:

1. areas considered to be inherently high risk, but believed to be well controlled
In these areas the organisation has established key processes and control systems that are expected to be adequate to manage the risks. It is important to provide periodic assessment to ensure processes and control systems are working as intended and to determine whether a more economic, efficient, or effective risk management regime can be put in place; and
2. areas where the residual risk is high
In these areas the internal audit is able to assist in the identification of processes and control regimes that will improve management of those risks.

Key processes and controls, and related internal audit activities for Country Energy's risks as identified in the risk profiles, were identified during interviews with stakeholders including the Executive team at Country Energy.

2. Network Management

Audits related to the Network business included:

- **Governance over Intelligence Network Strategy**

This audit focused on project governance over the program of works for the Intelligent Network Program against the Country Energy *Impact Project Methodology*, the requirements of the Project Management Office and KPMG's project management methodology.

The audit resulted in five key findings; four with a moderate priority rating and one with a low rating.

- **Network Risk Management System**

This audit focused on the effectiveness of current processes to manage operational risks faced by Networks and Infrastructure. The project mapped existing operational risk management processes against the key principals and elements of the Joint Australian/New Zealand Standard AS/NZ 4360:2004, *Risk Management*.

The audit resulted in only two key findings, both with a low priority rating.

- **Regional Centres**

This audit focused on:

- the processes and controls established to ensure implementation of the 'Service Delivery Consistency Reference' guidelines issued by Infrastructure Operations to drive consistency in processes across the regions
- whether key processes and associated controls relating to Safety (as required by the Consistency Reference), Budgeting and reporting, Field Services, Fleet, Purchasing, and Inventory management and warehouse security in the Far North Coast and North Western Regional Centres are operating effectively to mitigate identified risks to an acceptable level.

The audit resulted in twelve key findings; one high priority, six moderate and five low priority findings.

- **IT Disaster Recovery (phase II)**

Due to timing issues this internal audit was not completed in the 2008/2009 audit program as planned, but was included in the 2009/2010 audit program and is currently being finalised.

Action Plans to Address Audit Findings

Accountable managers have developed Agreed Management Actions to address the audit findings. The agreed actions are documented and monitored by an independent department within Country Energy. The performance against the actions is reported to the Audit and Risk Committee.

- **Safety Audits**

During 2008/2009 Country Energy engaged NCS International Pty Ltd to conduct two independent surveillance audits of our Occupational Health Safety and Environment Management System (OHSEMS), and achieved continued certification to AS/NZS 4801:2001 *Occupational Health and Safety Management Systems* and ISO 14001:2004 *guideline for an Environmental Management System*.

2. Network Management

2.3 INDEPENDENT APPRAISAL PROCESS

Country Energy engaged the services of Sinclair Knight Merz Pty Ltd to conduct an independent appraisal of this Network Performance Report 2008/2009. The appraisal has been conducted with due regard to the requirements of the Electricity Supply (Safety and Network Management) Regulation as well as the Independent Appraisal Guidelines given in the Electricity Network Performance Report Outline issued by Industry and Investment NSW.

2.4 NETWORK COMPLAINTS

Complaint information is taken directly from our customers either in person, over the phone, from our website contact form or from written complaints received. All complaints including those resolved at first contact are entered into Country Energy's dedicated Contact Management System (CMS).

The complaint is then assigned to the most appropriate actioning officer, depending on the category of the complaint, and an automatic escalation and reminder system ensures the complaint is dealt with in a timely manner.

All investigation and contact with the customer is recorded and contained within the CMS database.

Our system collects various kinds of data in separated fields within the complaint database, which can later be exported and sorted to provide useful statistics and trends, as well as data for regulatory reporting. Data is checked manually for integrity.

The CMS is currently being reviewed to ensure it is meeting business requirements. A number of additional data fields have been identified that could provide valuable information to assist with identifying and addressing systemic issues.

For example, the addition of a field within the Guaranteed Customer Service Standards (GCSS) form would allow the capture of the Field Service Centre responsible for a planned outage. Having this information would then allow us to report the number of GCSS breaches resulting from individual Field Service Centres. Area managers could then address the issue directly with Field Service Centre staff with a view to improving processes and achieving a better result.

The total network complaints trend continues to increase due to higher community expectations in reliability and quality of supply. Improvements in the process of recording complaint information have also contributed to the increase.

The increased trend in complaints relating to voltage issues and level of supply capacity could be in part attributed to the introduction of more sensitive and less tolerant electronic equipment, and to availability and affordability of split system air conditioning units installed without consideration of existing supply limitations.

The declining trend of complaints in the reliability sector could be attributed to the increased installation of network protection sites resulting in reduced extended supply failures and reduced duration due to smaller sectionalising of faulted area. This in turn could be also contributing to the increased trend in complaints associated with short duration interruptions (No. of < 1 min interruptions).

Table 2.1 - Complaints Received Regarding Network Issues					
	Previous Years				Current Year
	2004/2005	2005/2006	2006/2007	2007/2008	2008/2009
Complaints Total	2100	2541	2046	2327	3232
Complaints per 1,000 Distribution Customers	2.9	3.3	2.7	3.0	4.2
Complaints regarding Vegetation Management	133	105	152	164	169

2. Network Management

Table 2.2 – Network Complaint Investigations Completed 2008/2009

Category	Nature of Complaint	Number	Number Valid*
Voltage	Sustained over voltage	113	75
	Sustained under voltage	150	99
	Voltage fluctuations	291	143
	Voltage dips	78	36
	Switching transients	2	0
	N-E voltage difference	24	9
	Ground fault voltage	2	0
	Voltage unbalance	8	6
	Mains signalling voltages (Outside defined range)	0	0
	HV injection (HV/LV Intermix)	0	0
	Notching	0	0
	Invalid	300	
	Sub total (Supply Voltage Complaints)	668	368
Current	Direct current	0	0
	Harmonic content	0	0
	Inter Harmonics	0	0
	Invalid	0	0
	Sub total (Supply Current Complaints)	0	0
Other Quality	Mains signalling reliability	2	1
	Noise & Interference	144	63
	Level of supply capacity	74	41
	Supply frequency	0	0
	Level of EMF	2	1
	Invalid	116	
	Sub total (Other Quality of Supply Complaints)	222	106
	Sub total (All Quality of Supply Complaints)	890	474
Reliability	No. of supply failures	80	37
	Duration of supply failures	4	2
	No. of <1 min. interruptions	191	108
	Invalid	128	
	Sub total (Reliability of Supply)	275	147
Safety	Overhead line safety	8	5
	Underground safety	5	5
	Electrical station safety	0	0
	Service line safety	0	0
	Invalid	3	
	Sub total (Network Safety)	13	10
	Total	1178	631

* A complaint is valid where non-compliance with published service and network standards occur

2. Network Management

2.5 CUSTOMER SERVICE STANDARDS REPORTING

Table 2.3 – Customer Service Standards 2008/2009 Data				
	Payments Given Based on Interruption <u>Duration</u> (Total Number)	Payments Given Based on Interruption <u>Frequency</u> (Total Number)	Claims Not Paid Based on Interruption <u>Duration</u> (Total Number)	Claims Not Paid Based on Interruption <u>Frequency</u> (Total Number)
Metropolitan	NA	NA	NA	NA
Non-Metropolitan	2	0	17	12

Twelve of the claims not paid based on interruption duration are due to severe weather and the other five are due to not exceeding the 18 hour duration.

Five claims not paid based on interruption frequency are due to not having any eligible outages, whilst the remainder (7) had less than the required number of eligible outages.

3. Network Planning

3.1 OVERVIEW

Country Energy is committed to providing a safe, secure and reliable supply of energy in a cost effective manner. We achieve this by conducting network planning and making investment decisions in line with our CEK8018 *Network Asset Management Plan (NAMP)*.

The *NAMP* outlines Country Energy's obligations in relation to network planning, including the need to compile and publish an annual *Electricity System Development Review (ESDR)*. It has been prepared in accordance with the compliance obligations of network management legislation, regulations and related codes of practice, and reflects recognised industry best practices and standards relating to the management of electricity infrastructure assets. It is designed to conform with the NSW Government policy and planning guidelines on total asset management, which addresses strategic planning relating to capital investments, renewal, and maintenance.

Country Energy's network development is in accordance with the *Electricity Supply Act*, the *National Electricity Code*, the *NSW Code of Practice - Demand Management for Electricity Distributors* and our policy CEPG8003 *Subtransmission and Distribution Network Planning Criteria and Guidelines*.

In general, Country Energy plans the development of its network to ensure:

- network capacity is adequate to meet power transfer requirements
- electrical and thermal design ratings (normal and overload) of equipment are not exceeded
- supply reliability is in accordance with published standards, or as negotiated to meet the special requirements of individual major network customers
- quality of supply meets published standards and system voltage levels are maintained within acceptable standard limits
- safety standards are maintained or exceeded
- environmental constraints are satisfied, and
- the above requirements are met in a cost effective manner.

3.2 DESIGN PLANNING CRITERIA

In August 2005 the Minister for Energy and Utilities introduced new *licence conditions* that included requirements for planning and design of subtransmission and distribution networks.

Following a review of the *licence conditions* conducted by the Minister, those conditions were replaced with updated and revised conditions with effect from 1 December 2007.

The *design planning criteria* set out:

- input standards to be used by a licence holder in planning its network, and
- requirements for load forecasting and contingency planning methodologies intended to achieve operational outcomes.

The baseline levels of planned redundancy required under the *design planning criteria* underpin Country Energy's plans for the network to ensure, as far as is reasonably practicable, that:

- the *reliability standards* are met, and
- an adequate supply, with an appropriate level of redundancy, consistent with its regulatory obligations, is provided.

The *design planning criteria* applicable to Country Energy are listed in Table 3.1.

3. Network Planning

Table 3.1 - Design Planning Criteria

Network Element	Load Type	Forecast Demand or Expected Demand	Security Standard	Customer Interruption Time
Sub Transmission Line	Urban & Non-Urban	≥ 15 MVA	N-1 ¹	< 1 minute
	Urban & Non-Urban	< 15 MVA	N ²	Best practice repair time
Sub Transmission Substation	Urban & Non-Urban	Any	N-1	< 1 minute
Zone Substation	Urban & Non-Urban	≥ 15 MVA	N-1 ¹	< 1 minute
	Urban & Non-Urban	< 15 MVA	N ²	Best practice repair time
Distribution Feeder	Urban (regional centres) ⁴	Any	N-1 ³	< 4 Hours
	Urban (other) & Non-Urban	Any	N	Best practice repair time
Distribution Substation	Urban & Non-Urban	Any	N	Best practice repair time

1. For a *Subtransmission line - Overhead* and a Zone Substation:
 - a. Under N-1 conditions, the *forecast demand* is not to exceed the *thermal capacity* for more than 1 per cent of the time i.e. a total aggregate time of 88 hours per annum, up to a maximum of 20 per cent above the *thermal capacity* under N-1 conditions. For Country Energy, in other than regional centres, the *forecast demand* must not exceed the *thermal capacity* under N-1 conditions.
 - b. Under N conditions, a further criterion is that the *thermal capacity* is required to meet at least 115 per cent of forecast demand.

For a *Subtransmission line - Underground*, any overhead section may be designed as if it was a *Subtransmission line - Overhead*, providing the *forecast demand* does not exceed the *thermal capacity* of the underground section at any time under N-1 conditions.

2. Under N conditions, *thermal capacity* is to be provided for greater than 115 per cent of *forecast demand*.
3. By 30 June 2014, expected demand is to be no more than 80 per cent of feeder thermal capacity (under system normal operating conditions) with switchable interconnection to adjacent feeders enabling restoration for an unplanned network element failure. By 30 June 2019, expected demand is to be no more than 75 per cent of feeder thermal capacity. In order to achieve compliance, feeder reinforcement projects may need to be undertaken over more than one *regulatory period*. In those cases where a number of feeders form an interrelated system (such as a meshed network), the limits apply to the average loading of the feeders within the one system.
4. Regional centre means: Albury, Armidale, Ballina, Bathurst, Broken Hill, Coffs Harbour (including Sawtell), Dubbo, Forster-Tuncurry, Goulburn, Grafton, Griffith, Lismore, Orange, Port Macquarie, Queanbeyan, Tamworth, Taree, Tweed Heads, and Wagga Wagga.

3. Network Planning

3.2.1 DESIGN PLANNING CRITERIA COMPLIANCE REPORTING

Subtransmission Planning

The planning criterion considers the adequacy of the subtransmission network not only to meet the energy demand requirement, but also its capability to do so within component electrical and thermal ratings and voltage limits. Analysis is carried out to generally achieve an economically efficient outcome in which the subtransmission network is secure and within rating following the forced outage of any single circuit line or substation element during peak periods.

At reasonably high demand levels, but not at peak levels, consideration is given to generally ensuring the security of the network following an unforced outage. Line thermal ratings are based on a continuous rating and a short time rating relative to conductor design temperature.

Provision is also made for use of the short-time capability or cyclic rating of transformers during a planned outage of a parallel transformer.

Other subtransmission planning criteria outlines that:

- network equipment is designed to withstand maximum fault duty.
- any subtransmission fault is seen by at least two protection systems that are fully independent.

Each potential augmentation project is treated on its own merits. The main economic impact of subtransmission works is on the value of unsupplied energy, time to capacity limits being exceeded and the initial capital costs. These are quantified over the expected life of the project and costed at appropriate rates.

Planning of the subtransmission network is usually carried out over a longer time frame of between five to ten years. Country Energy distinguishes between long range and short range network planning, with long-term network planning generally confined to 132 and 66 kV networks with a horizon of five to ten years or beyond. Short-term network planning is usually limited to networks ranging from 11 to 33 kV, with a horizon period of up to five years.

Primary Distribution Planning

Country Energy's current distribution network planning and capital investment framework is driven more by technical drivers than pure business drivers. This is primarily due to the inherent geographical disabilities, our commitment to providing customers with published standards of reliability and quality of supply, and low profit structure of the network which is essentially radial.

Distribution network planning is generally of a short-term nature – up to five years. Country Energy's traditional approach to distribution network planning has been to focus on feeder augmentation works to provide supply to new or growing loads, maintain adequate voltage regulation, enhance reliability of supply, upgrade thermal capacity (and minimise system losses). This form of distribution planning criteria will continue.

Network Elements Not Complying with the Design Planning Criteria on 1 July 2008

For any new network element where planning commenced after 1 December 2007, Country Energy is required to comply with the design planning criteria. Plans have been developed to ensure that existing elements detailed in Tables 3.2 to 3.4 will be compliant by 1 July 2014 when the licence conditions apply to these elements.

3. Network Planning

Table 3.2 - Subtransmission Lines and Substations and Zone Substations		
Each Network Element or Class of Network Elements		
Not Complying with the Design Planning Criteria on 1 July of the Current Year (2008/2009)		
Element including Location, Customer Numbers, Element Length/Capacity	Description of Non-Compliance and Reason	Proposed Remedial Actions & Timetable
Forbes - West Jemalong 66kV feeder (37km) 7,000 customers Central Western Region	N-1 $\geq$ 15MVA subtransmission feeder. Radial 66kV feeder with peak demand of 15MVA, outage of 66kV feeder leads to loss of supply to all customers	Construct a second Forbes - West Jemalong 66kV feeder, expected completion 2011/12
TransGrid-Orange South-Orange West 66kV feeders (20km) 12,000 customers Central Western Region	N-1 $\geq$ 15MVA subtransmission feeder. Orange South and Orange West both have peak demands above 15MVA, 66kV feeders cannot be restored within 1 minute during contingency	Augment Orange South and Orange West zone substation, add 66kV feeder bays, expected completion 2011/12
Bathurst Russell St zone substation 8,000 customers Central Western Region	N-1 $\geq$ 15MVA zone substation. Russell St zone substation has a firm transformer capacity of 20MVA with a peak demand of 26MVA	Augment Russell St zone substation to 30MVA capacity, transfer load to adjacent zone substation, expected completion 2010/11
TransGrid - Parkes 66kV feeder (12km) 7,000 customers Central Western Region	N-1 $\geq$ 15MVA subtransmission feeder. 66kV feeder has peak demand of 20MVA. Outage of 66kV feeder results in overload of remaining in-service 66kV feeders	Construct a second 66kV feeder to Parkes Town, expected completion 2011/12
Temora - Thanowring 66kV feeder (8km) 7,000 customers Central Western Region	N-1 $\geq$ 15MVA subtransmission feeder. Radial 66kV feeder with peak demand of 15MVA, outage of 66kV feeder leads to loss of supply to all customers	Construct a second Temora - Thanowring 66kV feeder, expected completion 2012/13
Koolkhan - Maclean 66kV feeder (36km) 11,000 customers Far North Coast Region	N-1 $\geq$ 15MVA subtransmission feeder. Radial 66kV feeder with peak demand above 20MVA, outage of 66kV feeder leads to loss of supply to all customers	Construct a second Koolkhan - MacLean 66kV feeder, expected completion 2010/11
Lismore 132kV - Lismore South 66kV feeders (2x3km) 17,000 customers Far North Coast Region	N-1 $\geq$ 15MVA subtransmission feeder. 66kV network to Lismore South has peak demand of 35+MVA. Outage of 66kV feeder results in overload of remaining in-service 66kV feeder	Augment Lismore 132kV - Lismore South 66kV feeders, expected completion 2009/10
Ewingsdale - Mullumbimby 66kV feeder (13km) 20,000 customers Far North Coast Region	N-1 $\geq$ 15MVA subtransmission feeder. Outage of 66kV feeder results in voltage levels below standard and overload of remaining in-service 66kV feeders	Staged conversion of 66kV network to 132kV, expected completion of Stage 1 in 2012/13
Terranora transmission substation 35,000 customers Far North Coast Region	N-1 $\geq$ 15MVA transmission substation. Terranora transmission substation has a firm transformer capacity of 100MVA with a peak demand of 110MVA	Install a third 110/66kV transformer at Terranora, expected completion 2009/10
Merbein - Dareton 66kV feeder (Buronga load) Far West Region	N-1 $\geq$ 15MVA subtransmission feeder. 66kV radial feeder with a peak demand of 30+MVA. Outage of 66kV feeder leads to loss of supply to all customers	Establish Buronga 66/22kV zone substation, construct 66kV feeder from Gol Gol to Buronga, reconfigure 22kV feeder to 66kV, expected completion 2010/11
Coffs Harbour North zone substation 10,000 customers Mid North Coast Region	N-1 $\geq$ 15MVA zone substation. Coffs Harbour North zone substation has a firm transformer capacity of 23MVA with a peak demand of 30MVA	Augment Coffs Harbour North zone substation to 45MVA capacity, expected completion 2009/10
Karangie - Coffs Harbour North 66kV feeder (11km)	N-1 $\geq$ 15MVA subtransmission feeder. 66kV feeder has peak demand of 30MVA. Outage of 66kV feeder results	Construct a Coffs Harbour North - Coffs Harbour South 66kV feeder, expected

3. Network Planning

Table 3.2 - Subtransmission Lines and Substations and Zone Substations		
Each Network Element or Class of Network Elements		
Not Complying with the Design Planning Criteria on 1 July of the Current Year (2008/2009)		
Element including Location, Customer Numbers, Element Length/Capacity	Description of Non-Compliance and Reason	Proposed Remedial Actions & Timetable
10,000 customers Mid North Coast Region	in overload of remaining in-service 66kV feeders and extreme low voltage levels	completion 2011/12
Coffs Harbour South zone substation 9,000 customers Mid North Coast Region	N-1 $\geq$ 15MVA zone substation. Coffs Harbour South zone substation has a firm transformer capacity of 23MVA with a peak demand of 30MVA	Establish a 132/66/11kV transmission substation at Boambee South in joint planning with TransGrid's constraints, transfer load away from Coffs Harbour South zone substation, expected completion 2009/10
Raleigh zone substation 6,000 customers Mid North Coast Region	Raleigh zone substation has a firm transformer capacity of 10MVA with a forecast peak demand of 13MVA in 2013/14.	Establish a 132/33/11kV transmission substation at Raleigh in joint planning with TransGrid's 132kV network constraints, expected completion 2009/10
Newee Creek zone substation 4,000 customers Mid North Coast Region	Newee Creek zone substation has a firm transformer capacity of 12MVA with a forecast peak demand of 11MVA in 2013/14	Establish a 132/11kV transmission substation at Macksville in joint planning with TransGrid's 132kV network constraints, expected completion 2010/11
Boggy Creek - Nambucca Heads 66kV feeder (4km) 6,000 customers Mid North Coast Region	N-1 $\geq$ 15MVA subtransmission feeder. Radial 66kV feeder with peak demand of 15MVA in 2014, outage of 66kV feeder leads to loss of supply to all customers	Construct a second Boggy Creek to Nambucca 66kV feeder, expected completion 2013/14
Stroud - Gloucester 33kV feeders (2 x 40km) 4,000 customers Mid North Coast Region	N-1 $\geq$ 15MVA subtransmission feeder. 33kV feeders have peak demand of 15MVA. Outage of 33kV feeder results in voltage levels below standard and overload of remaining in-service 33kV feeder	Establish a 33kV switching station at Wards River, expected completion 2009/10
Taree - Kew 66kV feeder (45km) 7,000 customers Mid North Coast Region	N-1 $\geq$ 15MVA subtransmission feeder. 66kV feeder has peak demand of 15MVA. Outage of 66kV feeder results in voltage levels below standard and overload of remaining in-service 66kV feeder	Establish a 132/66/11kV transmission substation at Herons Creek, expected completion 2012/13
Taree - Failford - Tuncurry - Forster - Bohnock 66kV feeders (100km) 18,000 customers Mid North Coast Region	N-1 $\geq$ 15MVA subtransmission feeder. 66kV feeders have peak demand of 40+MVA. Outage of 66kV feeder results in voltage levels below standard and overload of remaining in-service 66kV feeders	Establish a 132/66kV transmission substation at Nabiack and construct interconnecting 66kV feeders, expected completion 2011/12
Forster - Tuncurry 66kV feeder (8km) 9,000 customers Mid North Coast Region	N-1 $\geq$ 15MVA subtransmission feeder. 66kV feeders have peak demand of 20MVA. Outage of 66kV feeder results in overload of remaining in-service 66kV feeder	Replace existing 66kV feeder river crossing with 66kV cable on bridge, expected completion 2009/10
TransGrid - Glen Innes 66kV feeder (9km) 7,000 customers Northern Region	N-1 $\geq$ 15MVA subtransmission feeder. 66kV feeder has peak demand of 15MVA. Limited capacity of remaining in-service 66kV feeder	Construct a second TransGrid - Glen Innes 66kV feeder in joint planning with TransGrid 132kV constraints, expected completion 2009/10
TransGrid - Gunnedah 66kV feeders (2 x 12km) 7,000 customers Northern Region	N-1 $\geq$ 15MVA subtransmission feeder. 66kV feeders have peak demand of 23MVA. Outage of 66kV feeder results in overload of remaining in-service 66kV feeder	Upgrade TransGrid - Gunnedah 66kV feeders, expected completion 2009/10
Borthwick St - Wynne St 66kV feeder,	N-1 $\geq$ 15MVA subtransmission feeder and N-1 $\geq$ 15MVA zone substation. Wynne St substation with a peak	Augment Wynne St and Borthwick St substations and loop in/out Ashford

3. Network Planning

Table 3.2 - Subtransmission Lines and Substations and Zone Substations		
Each Network Element or Class of Network Elements		
Not Complying with the Design Planning Criteria on 1 July of the Current Year (2008/2009)		
Element including Location, Customer Numbers, Element Length/Capacity	Description of Non-Compliance and Reason	Proposed Remedial Actions & Timetable
Wynne St zone substation 8,000 customers Northern Region	demand of 25MVA is controlled by single 66kV circuit breaker at Borthwick St. Transformer or feeder fault leads to loss of supply to all customers with restoration greater than 1 minute	66kV feeder to Wynne St, expected completion 2012/13
TransGrid - South Tamworth - Oxley Vale 66kV feeders (25km) 11,000 customers Northern Region	N-1 $\geq$ 15MVA subtransmission feeder. 66kV feeders have peak demand of 40MVA. Outage of 66kV feeder results in overload of remaining in-service 66kV feeders and supply cannot be restored in 1 minute	Augment TransGrid - South Tamworth - Oxley Vale feeders and configure substation to 66kV 'ring' network, expected completion 2010/11
TransGrid - Goddard Lane 66kV feeder (10km) 5,000 customers Northern Region	N-1 $\geq$ 15MVA subtransmission feeder. Radial 66kV feeder with peak demand above 15MVA, outage of 66kV feeder leads to loss of supply to all customers	Construct a second TransGrid - Goddard Lane 66kV feeder, expected completion 2010/11
TransGrid - Quirindi 66kV feeder (56km) 7,000 customers Northern Region	N-1 $\geq$ 15MVA subtransmission feeder. Radial 66kV feeder with peak demand of 15MVA, outage of 66kV feeder leads to loss of supply to all customers	Construct a second TransGrid - Quirindi 66kV feeder, expected completion 2011/12
Beryl - Mudgee (tee) 132kV feeder (35km) 9,000 customers North Western Region	N-1 $\geq$ 15MVA subtransmission feeder. 132kV feeder has peak demand of 25MVA. Outage of 132kV feeder results in overload of remaining in-service 66kV feeder	Augment Beryl - Mudgee subtransmission network, expected completion 2012/13
Beryl - Dunedoo 66kV feeder (41km) 7,000 customers North Western Region	N-1 $\geq$ 15MVA subtransmission feeder. Radial 66kV feeder with peak demand above 15MVA, outage of 66kV feeder leads to loss of supply to all customers	Construct a second Beryl - Dunedoo 66kV feeder, expected completion 2012/13
Nyngan - Cobar 66kV feeder (130km) 4,000 customers North Western Region	N subtransmission feeder. Radial 66kV feeder with voltage levels below standard during system normal operation	Install a 66kV regulator on Nyngan - Cobar 66kV feeder, expected completion 2010/11
Dubbo - Yarrandale 66kV feeder (5.5km) 9,000 customers North Western Region	N-1 $\geq$ 15MVA subtransmission feeder. Radial 66kV feeder with peak demand above 15MVA, outage of 66kV feeder leads to loss of supply to all customers	Construct a second Dubbo - Yarrandale 66kV feeder, expected completion 2010/11
Yarrandale - Gilgandra 66kV feeder (65km) 5,000 customers North Western Region	N-1 $\geq$ 15MVA subtransmission feeder. Radial 66kV feeder with peak demand above 15MVA, outage of 66kV feeder leads to loss of supply to all customers	Construct a second Yarrandale - Gilgandra 66kV feeder, expected completion 2014/15
Wellington - Dubbo 132kV feeders (2 x 47km) 70,000 customers North Western Region	N-1 $\geq$ 15MVA transmission feeder. 132kV feeder with peak demand above 150MVA, outage of a 132kV feeder leads to overload of remaining in-service 132kV feeder	Construct a Wellington - Narromine 132kV feeder, expected completion 2012/13
Narromine - Dubbo West subtransmission network 6,000 customers North Western Region	N-1 $\geq$ 15MVA zone substation. Dubbo West approaching 15MVA, currently supplied via a radial from Dubbo 132 to West Dubbo and to Narromine, provides N-1 by a 66kV supply at Narromine, via the establishment of a 132/66kV transformer. A switched 132kV busbar will provide a connection point for a future Wellington - Narromine 132kV feeder	Establish a 132/66kV transmission substation at Narromine 132kV tee, expected completion 2010/11
Yarrandale zone substation	Yarrandale has a single 10MVA transformer with peak	Augment Yarrandale zone substation,

3. Network Planning

Table 3.2 - Subtransmission Lines and Substations and Zone Substations		
Each Network Element or Class of Network Elements		
Not Complying with the Design Planning Criteria on 1 July of the Current Year (2008/2009)		
Element including Location, Customer Numbers, Element Length/Capacity	Description of Non-Compliance and Reason	Proposed Remedial Actions & Timetable
4,000 customers North Western Region	demand approaching 10MVA, and expected additional load, a larger transformer and another transformer is required. A new second 66kV feeder from Dubbo 132kV to Yarrandale requires 66kV feeder bays at Yarrandale.	expected completion 2010/11
Bega - Pambula - Eden South 66kV feeder (60km) South Eastern Region	N-1 $\geq$ 15MVA subtransmission feeder. 66kV radial feeder with a peak demand of 20+MVA. Outage of 66kV feeder leads to loss of supply to all customers.	Construct second Bega - Eden South 66kV feeder over redundant 33kV easement, expected completion 2011/12
Cooma - Bega 132kV feeder (92km) and Cooma - Bombala - Bega 66kV feeder (96km) 23,000 customers South Eastern Region	N-1 $\geq$ 15MVA subtransmission feeders. 132kV and 66kV network with peak demand of 50MVA. Outage of 132kV feeder leads to overload of 66kV feeder, results in loss load of 30+MVA	Construct a Cooma - Bega dual 132/66kV feeder on existing 66kV easement, expected completion 2010/11
Evans Lane - Batemans Bay 132kV feeder (57km) 25,000 customers South Eastern Region	N-1 $\geq$ 15MVA subtransmission feeder. 132kV network with peak demand of 50+MVA. Outage of 132kV feeder leads to extreme low voltage levels and possible voltage collapse.	Rebuild smaller 132kV feeder with larger conductor and staged power factor correction, expected completion 2012/13. (Project may change dependant on projects in Integral Energy 132kV network)
Mulwala - Finley 132kV and 66kV feeders (60km) 7,000 customers Southern Region	N-1 $\geq$ 15MVA subtransmission feeders. Both Finley zone substation and Mulwala 132/66kV substation have demands above 15MVA. On loss of the 132kV feeder to Mulwala and the 66kV feeder to Finley, the remaining in-service feeder is overloaded.	Establish a 132/22kV substation at Mulwala, construct a 132kV feeder from Mulwala tee to new 132/22kV substation, construct 132/66kV dual feeder from Finley 132kV to Finley zone substation, convert 66kV Finley - Mulwala feeder to 132kV, estimate completion 2010/11
Griffith 33kV feeders (37km) 12,000 customers Southern Region	N-1 $\geq$ 15MVA subtransmission feeders. 33kV feeders have peak demand above 15MVA. Outage of 33kV feeder results in overload of remaining in-service 33kV feeders.	Initially construct 132kV feeder operated at 33kV. Establish a Griffith West 132/33kV transmission substation and 33kV feeder interconnections into the long term, expected initial feeder completion 2012/13
Deniliquin to Moama 66kV feeder (70km) 4,000 customers Southern Region	N-1 $\geq$ 15MVA subtransmission feeder. 66kV radial feeder with a peak demand of 15+MVA. Outage of 66kV feeder leads to loss of supply to all customers	Construct a second Deniliquin to Moama 66kV feeder, expected completion 2011/12
Ashmont zone substation 15,000 customers (Wagga City) Southern Region	N-1 $\geq$ 15MVA subtransmission feeders. The Wagga Wagga city 66kV network is arranged in a 'tee' step ladder formation, with some zone substations not having incoming 66kV feeder or 66kV transformer protection. Up to four zone substations can be affected by a single feeder or transformer fault. Restoration of supply within 1 minute is not possible.	Augment Ashmont substation to fully switched 66kV substation and construct 66kV feeder connection, expected completion 2009/10
Wagga - Temora 132kV feeder (80km) 9,000 customers Southern Region	N-1 $\geq$ 15MVA subtransmission feeder. 132kV feeder with limited 66kV backup, Peak demand of 50MVA. Outage of 132kV feeder leads to overload of 66kV feeder and results in loss load of 20+MVA	Construct a second Wagga to Temora 132kV feeder, expected completion 2012/13

3. Network Planning

Table 3.2 - Subtransmission Lines and Substations and Zone Substations		
Each Network Element or Class of Network Elements		
Not Complying with the Design Planning Criteria on 1 July of the Current Year (2008/2009)		
Element including Location, Customer Numbers, Element Length/Capacity	Description of Non-Compliance and Reason	Proposed Remedial Actions & Timetable
Wagga - Lockhart - Kywong - Narrandera - Yanco 66kV feeder (85km) 5,000 customers Southern Region	N-1 $\geq$ 15MVA subtransmission feeder. 66kV backup to Narrandera limited by low voltage levels.	Install a 66kV regulator on Wagga - Narrandera 66kV feeder near Lockhart, expected completion 2009/10
Bourkelands zone substation 15,000 customers (Wagga City) Southern Region	N-1 $\geq$ 15MVA subtransmission feeders. The Wagga city 66kV network is arranged in a 'tee' step ladder formation, with some zone substations not having incoming 66kV feeder or 66kV transformer protection. Up to four zone substations can be affected by a single feeder or transformer fault. Restoration of supply within 1 minute is not possible.	Augment Bourkelands substation with fully switched in/out 66kV busbar, expected completion 2010/11
Bomen zone substation 15,000 customers (Wagga City) Southern Region	N-1 $\geq$ 15MVA subtransmission feeders. The Wagga city 66kV network is arranged in a 'tee' step ladder formation, with some zone substations not having incoming 66kV feeder or 66kV transformer protection. Up to four zone substations can be affected by a single feeder or transformer fault. Restoration of supply within 1 minute is not possible.	Augment Bomen substation, expected completion 2009/10
Cartwrights Hill zone substation 15,000 customers (Wagga City) Southern Region	N-1 $\geq$ 15MVA subtransmission feeders. The Wagga city 66kV network is arranged in a 'tee' step ladder formation, with some zone substations not having incoming 66kV feeder or 66kV transformer protection. Up to four zone substations can be affected by a single feeder or transformer fault. Restoration of supply within 1 minute is not possible.	Augment Cartwrights Hill substation with fully switched in/out 66kV busbar, expected completion 2011/12
Wagga - Uranquinty 66kV feeder (19km) 7,000 customers Southern Region	N-1 $\geq$ 15MVA subtransmission feeder. Radial 66kV feeder with peak demand of 18MVA, outage of 66kV feeder leads to loss of supply to all customers	Construct Bourkelands - Uranquinty 66kV feeder and augment Uranquinty 66kV busbar, expected completion 2011/12
Wagga Wagga City Network 66kV feeders (5km) 15,000 customers (Wagga City) Southern Region	N-1 $\geq$ 15MVA subtransmission feeders. The Wagga Wagga city 66kV network is arranged in a 'tee' step ladder formation, with some zone substations not having incoming 66kV feeder or 66kV transformer protection. Up to four zone substations can be affected by a single feeder or transformer fault. Restoration of supply within 1 minute is not possible.	Construct a number of short 66kV feeders between zone substations at Ashmont, Bomen and Cartwrights Hill, expected completion 2011/12

3. Network Planning

Table 3.3: Distribution Feeder Summary Report by Class of Network Elements

Not Complying with the Design Planning Criteria on 1 July of the Current Year (2008/2009)

N-1 Regional Centres

Total Number of Feeders	Number of Feeders Without N-1 Capability (1 Minute)	Description and Reason for Non-Compliance	Proposed Remedial Actions & Timetable
351	157	Insufficient feeder thermal ratings, lack of interconnectors, underrated interconnecting switchgear	Upgraded and additional feeders, construction of interconnectors, upgrading and additional interconnecting switchgear (gas switches, reclosers) N-1 programs have been set and are scheduled to be completed by 2014. There are 518 programs of work scheduled, of which 15% have been completed.

URBAN⁴

Total Number of Feeders	Number of Feeders Without N Capability	Description and Reason for Non-Compliance	Proposed Remedial Actions & Timetable
As above	As above	As above	As above

NON-URBAN

Total Number of Feeders	Number of Feeders Without N Capability	Description and Reason for Non-Compliance	Proposed Remedial Actions & Timetable
1026	Nil	Nil	As N capability is exceeded, remedial plans will be developed accordingly.

Table 3.4: Distribution Substation Summary Report by Class of Network Elements

Not Complying with the Design Planning Criteria on 1 July of the Current Year (2008/2009)

CBD

Total Number of Substations	Number of Substations Without N-1 Capability (1 Minute)	Description and Reason for Non-Compliance	Proposed Remedial Actions & Timetable
Not Applicable	Not Applicable	Not Applicable	Not Applicable

URBAN and NON-URBAN

Total Number of Substations	Number of Substations Without N Capability	Description and Reason for Non-Compliance	Proposed Remedial Actions & Timetable
131,482	Nil	Nil	As N capability is exceeded, remedial plans will be developed accordingly.

⁴ By definition, urban for Country Energy means areas within a regional centre (page 8 of the Licence Conditions 2007).

3. Network Planning

3.3 DEMAND MANAGEMENT

The key drivers for capital investment in Country Energy's subtransmission and distribution systems fall into the general categories of growth, renewal, safety and reliability. Country Energy's process for scoping major capital projects includes identifying issues in all of these categories and, consequently, many of the major capital works projects initiated within Country Energy have multiple components.

Much of the capital investment in Country Energy's network is driven by N-1 reliability issues on radial subtransmission lines. This requires a demand management solution that can fully support the load supplied at all times (i.e. the equivalent of a second line) and effectively limits the non network options for embedded base load generation. The market has not been able to provide a reliable, economic alternative for this class of constraint.

Another significant factor in capital investment within Country Energy is the replacement of older equipment for safety and operational reasons. In these types of projects where augmentation of equipment is a secondary component achieved at incremental cost there is little opportunity to source a timely or cost effective non network alternative.

Country Energy's network demand growth is generally driven by population growth particularly along the coastal strips – in conjunction with an increase in individual demand due to lifestyle factors. Air conditioning has been a major contributor to an emerging summer peaking trend and this has proven difficult to address using conventional non-network options.

Country Energy seeks initiatives which can be applied to address specific network constraints or customer groups which include:

- embedded generation
- emerging technologies
- negotiated outcomes

There are comparatively few large customers or embedded generators in Country Energy's network which further limits opportunities for large scale load reductions in locations that are subject to network constraints. There are significant opportunities for Demand Management options which can be applied at a broader level to slow demand growth and address emerging distribution network constraints. This will require investment in the education of customers and the development of new, innovative technologies to support energy efficiency and small scale embedded generation initiatives. The Demand Management Innovation Allowance Scheme for the 2009-2014 regulatory period will be used to broaden our knowledge base in these areas in order to develop and promote non network alternatives for the longer term benefit of customers.

Facilitation of Renewable Energy Connection.

Country Energy recognises that distributed energy resources – particularly renewables – will become increasingly important in reducing power flows and losses in the distribution network. In order to minimise barriers to increased penetration of distributed energy resources metering upgrades for residential customers installing photovoltaic arrays are provided free of charge.

3. Network Planning

Demand Management Process

Country Energy has implemented internal procedures to comply with the Demand Management Code of Practice and National Electricity Rules. This ensures the network planning process is transparent, consultative and equitable in the treatment of Demand Management solutions relative to traditional network alternatives. This process includes:

- publication of the Annual Electricity System Development Review
- maintenance of a Register of Interested Parties
- review of emerging constraints with an expected network solution in excess of \$1 million.
- publication of Consultation Papers via AEMO and Country Energy external web page.
- notification to Interested Parties of Demand Management opportunities.
- consultation with prospective Demand Management Service Providers.

Demand Management Initiatives

Country Energy pro-actively seeks Demand Management solutions through provision of policy advice, monitoring of emerging and innovative technologies and participation in groups, including:

- Intelligent Grid Advisory Group (IGAG)
- ENA Demand Management and Embedded Generation Committee (DMEGC)
- CenDEP Centre for Distributed Energy and Power
- industry consultation with external parties to monitor and develop new technologies.

Intelligent Network

Country Energy is part of an international collaboration of utility companies looking at developing intelligent utility networks.

Convened by IBM, the Intelligent Utility Network Coalition aims to accelerate the research, development and deployment of intelligent network projects, dividing the tasks and sharing the gains. Country Energy is actively involved in the Embedded Generation working group of this coalition with the particular aim of facilitating utilisation of distributed renewable generation and energy storage technologies as an alternative to network augmentation.

Country Energy is the only Australian representative in this unique group of companies servicing more than six million customers across the world, collaborating on research and development programs.

Inverter Research and Development

Country Energy has formed an alliance with an Australian power electronics designer and manufacturer for the development of a grid interactive inverter to provide network support at low voltage and distribution level. A single phase, 240 volt, 20 kVA prototype has been designed and tested and three units are currently under construction to be installed at Country Energy's Research and Demonstration Centre in Queanbeyan together with three 20 kWh battery banks for field trials.

Unlike standard photovoltaic inverters which are one way and simply export the energy from solar cells as it is produced, a grid interactive inverter has "four quadrant" capability, allowing for bi-directional flow of both real (kW) and reactive (kVAr) power. This provides much greater flexibility for the grid interactive inverter in a variety of network based applications aimed at reducing network peak loads, saving network losses (plus associated CO₂), increasing network utilisation and improving quality of supply for customers.

3. Network Planning

The 20 kVA unit is modular and can be assembled in any single, two or three phase arrangement up to a few hundred kVA per phase with or without energy storage. It will effectively fill the gap between the small inverters installed in conjunction with renewable generation and much larger units which are generally purpose built for a particular application. Design innovations in switching frequency, control algorithms and assembly techniques have resulted, even at prototype stage, in substantial efficiency improvements and unit cost reductions relative to existing units (which have less functionality). Further gains are expected in production runs.

Many of Country Energy's rural feeders have long sections of SWER, or single phase steel, conductors where gradual load increases and large point loads impact severely on power supply quality. Standard network solutions would be to install voltage regulators or rebuild part of the line using a larger conductor, which can run into hundreds of thousands of dollars. An alternative is to use an inverter with a battery bank to store energy during light load periods and release it during peaks to reduce the load on the network. The inverter will also smooth out short term fluctuations during start up of large loads and reduce voltage rises in SWER systems caused by charging currents during light load periods.

In older low voltage overhead networks, the original designs were based on much lower household loads. However, lifestyle changes – such as increased penetration of air conditioners – can cause similar problems. The normal network response is either to establish a new distribution substation and reconfigure the low voltage or upgrade the low voltage conductors. In many cases, grid-interactive inverters providing reactive power to stabilise voltage levels are a very cost effective non-network alternative.

Other applications include:

Power factor correction – the grid interactive inverter can provide reactive power to adjust the power factor of the load on the network reducing network loads and losses. The benefits of an inverter over fixed capacitors for this application are that the output is continuously variable and independent of voltage so there are no step changes. The inverter response is instantaneous so the power factor can be constantly adjusted.

Short term fluctuations – while the steady state operation of the inverter is limited by thermal ratings the unit has significantly higher short term capability and can provide ride through support for dips and surges from sudden connection or disconnection of large loads.

Improved generation connection capacity – high level concentrations of embedded generation such as rooftop photovoltaics can create problems with voltage rise on the network. The grid interactive inverter can enhance the network's capacity to absorb generation by providing inductive reactive power to reduce voltage levels.

Ancillary benefits – other options to be explored further during field trials include local harmonic suppression and voltage balancing.

Conductor Replacement Program.

Country Energy has an ongoing program of replacing older conductors to maintain acceptable levels of reliability and safety. Rather than replacing like for like, a parallel initiative to augment conductors at incremental cost provides loss reductions, resulting in reduced peak demands and energy savings in the supply path.

3. Network Planning

Demand Management Program

Country Energy's Demand Management Program addresses key strategies in the delivery of Demand Management opportunities within Country Energy in the following key areas:-

- regulatory compliance
- program delivery
- awareness and education
- emerging technologies
- stakeholder communication

This program includes assessment of current projects through the Demand Management process and development of a Demand Management Program for the Regulatory Period 2009-2014. This includes both assessment of projects through the planning process as constraints emerge, which is supported by the extension of the D-Factor Scheme in NSW, and the more broadly based Demand Management Programs supported by the Demand Management Innovation Allowance Scheme commencing in the 2009/2010 financial year.

During 2008/2009, Demand Management assessments were carried out on 15 major projects. Of these, 14 were found to be non-viable due to the likely cost of demand reduction being significantly higher than the avoided cost of the augmentation component of the project.

One consultation paper was issued for the construction of a single 132 kV powerline from Mullumbimby to Ballina and an offer to develop a demand reduction option was received. At the end of June a feasibility study was in progress, with the study report due in mid July

3. Network Planning

Performance Data

Table 3.5 - Demand Management Projects Implemented During 2008/2009

	Description of Demand Management Project Implemented	Peak Demand Reduction (kVA)	CO2 Reduction (Tonnes Per Year and Expected Duration)	PV of Costs of Demand Management Project	PV of Total of Capital Expenditure Deferment plus Operating Expenditure Savings
Individual large projects					
1	Tweed Heads South 7.5 MVar capacitor bank	500	1,812 / 20 years	\$534,000	\$750,000
2	Oxley Vale 7.5 MVar capacitor bank	800	2,579 / 20 years	\$534,000	\$1,200,000
3	Ewingsdale 7.5 MVar capacitor bank	1,200	3,688 / 20 years	\$534,000	\$1,800,000
4	Lismore University 7.5 MVar capacitor bank	800	2,579 / 20 years	\$534,000	\$1,200,000
5	Lennox Head 7.5 MVar capacitor bank	250	806 / 20 years	\$534,000	\$375,000
Sub-total		3,550	11,464 / 20 years	\$2,670,000	\$5,325,000
Consolidated projects					
1	Various Conductor Upgrades throughout Country Energy's Distribution Network	6,877 est.	16,632 est. / 20 years	Incremental cost estimated at \$10,000/km 471km \$4,710,000	\$13,754,000
2	Load System Control Upgrades. Estimated 25% of total \$4,574,000 spend). Approx 20,000 relays Tweed Heads, Port Macquarie, Albury, Deniliquin, Inverell	Estimated 2,000 through improved control of off peak load switching	Not significant as only minor reduction in energy usage.	\$1,144,000	\$4,000,000
3	Capacitor banks	700	2,257 / 20 years	\$350,000	\$1,050,000
4	Facilitation of renewables connection through metering upgrades (net community benefit shown)	Est 1,915 (25% of 7,658kW installed capacity coincident with peak demand)	10,185 (based on 1.4 MWh per kW installed) / 10 years	\$1,444,728 (3,853 sites @\$376)	\$3,800,000
5	Negotiated Outcomes	Negotiations with rural customers to reduce large load connections such as underfloor heating methods on rural feeders			
6	Intelligent Network	Project in Development.			
Sub-total		11,492	29,074	\$7,648,728	\$22,604,000
Total		15,042	40,538	\$10,318,728	\$27,929,000

3. Network Planning

Table 3.6 – Demand Management Investigations Found Non-Viable in Current Year

	Description of Potential Demand Management Project Investigated and Reason for Non-viability	PV of Costs of Investigations ⁵
1	Mulwala 132/66kV Substation replacement The augmentation component is \$310K. It is not viable to find the required demand reduction of 3.75 MVA to avoid the annualised augmentation cost.	Internal investigation and reasonableness test cost is estimated at \$5,000.
2	Eden South transformer upgrade The augmentation component is \$90K. It is not viable to find the required demand reduction of 2.1 MVA to avoid the annualised augmentation cost.	Internal investigation and reasonableness test cost is estimated at \$10,000.
3	Stroud 33/11kV transformer replacement The augmentation component is \$212K. It is not viable to find the required demand reduction of 2.0 MVA to avoid the annualised augmentation cost.	Internal investigation and reasonableness test cost is estimated at \$5,000.
4	Bomen Substation augmentation The augmentation component is \$241K. It is not viable to find the required demand reduction of 2.0 MVA to avoid the annualised augmentation cost.	Internal investigation and reasonableness test cost is estimated at \$5,000.
5	Lismore South 66kV cables The augmentation component is \$7.6M. It is not viable to find the required demand reduction of 5.0 MVA to avoid the annualised augmentation cost.	Internal investigation and reasonableness test cost is estimated at \$10,000.
6	Alstonville to Ballina 66kV line upgrade The augmentation component is \$2.5M. It is not viable to find the required demand reduction of 13.0 MVA to avoid the annualised augmentation cost.	Internal investigation and reasonableness test cost is estimated at \$10,000.
7	Ashmont Substation augmentation The augmentation component is \$241K. It is not viable to find the required demand reduction of 2.0 MVA to avoid the annualised augmentation cost.	Internal investigation and reasonableness test cost is estimated at \$5,000.
8	Lockhart 66kV voltage regulator The augmentation component is \$4M. It is not viable to find the required demand reduction of 2.6 MVA to avoid the annualised augmentation cost.	Internal investigation and reasonableness test cost is estimated at \$15,000.
9	Oaks Estate Substation augmentation The augmentation component is \$1.2M. It is not viable to find the required demand reduction of 1.0 MVA to avoid the annualised augmentation cost.	Internal investigation and reasonableness test cost is estimated at \$10,000.
10	Madgwick Drive transformer upgrade The augmentation component is \$100K. It is not viable to find the required demand reduction of 2.0 MVA to avoid the annualised augmentation cost.	Internal investigation and reasonableness test cost is estimated at \$5,000.
11	Pacific Palms Substation augmentation The augmentation component is \$150K. It is not viable to find the required demand reduction of 0.3 MVA to avoid the annualised augmentation cost.	Internal investigation and reasonableness test cost is estimated at \$5,000.

⁵ Country Energy does not keep separate records of the cost of internal investigation.

3. Network Planning

Table 3.6 – Demand Management Investigations Found Non-Viable in Current Year

	Description of Potential Demand Management Project Investigated and Reason for Non-viability	PV of Costs of Investigations ⁵
12	Jindabyne east Zone Substation The augmentation component is \$7.3M. It is not viable to find the required demand reduction of 4.2 MVA to avoid the annualised augmentation cost.	Internal investigation and reasonableness test cost is estimated at \$10,000.
13	Narooma 33kV capacitor bank The augmentation component is \$1.5M. It is not viable to find the required demand reduction of 1.0 MVA to avoid the annualised augmentation cost.	Internal investigation and reasonableness test cost is estimated at \$15,000.
14	Smithtown to South West Rocks 66kV line The augmentation component is \$7.8M. It is not viable to find the required demand reduction of 8.0 MVA to avoid the annualised augmentation cost.	Internal investigation and reasonableness test cost is estimated at \$10,000.

4. Asset Management

4.1 OVERVIEW

Asset management is a critical component of Country Energy's overall network management strategy and has an important role in determining the outcomes for both the business and our customers.

The *NAMP* provides a framework for strategic management of our physical system assets to best support network service delivery. It includes our asset management strategies, policies, processes, resources, and our planned capital investments, asset maintenance and demand management. The *NAMP* is closely related to annual budgets and forecasts for capital, operating and maintenance expenditure planning.

The *NAMP*'s primary objectives include:

- establishing priorities in line with organisational objectives and statutory obligations, namely safety, reliability and sustainability
- planning and controlling financing and expenditure in accordance with these priorities, and
- ensuring resources are used as effectively and efficiently as possible so that the government and the community receive the most value for money.

These objectives cover the three major elements of asset management outlined in the *NAMP*:

- capital investment strategic planning
- asset renewal and replacement strategic planning, and
- asset maintenance strategic planning

The *NAMP* is designed to comply with the State Government's policy on Total Asset Management (TAM). It includes a five year forecast and reviews capital investment, refurbishment and asset maintenance strategies to ensure a focus on long term, system wide and whole of life management. These strategies ensure delivery of a secure, high quality, reliable and safe electricity network service that meets the needs and expectations of customers, community, shareholders, and other stakeholders at the lowest possible price, and complies with related statutory and regulatory requirements.

Country Energy reviews this plan annually.

4. Asset Management

4.2 TECHNICAL SERVICE STANDARDS

The Electricity Supply Standards adopted by Country Energy are set out in the document *CEPG8026 Electricity Supply Standard*, in accordance with the *Code of Practice – Electricity Service Standards* published by the former Electricity Association of NSW. The document outlines our objectives for technical performance of the network and ensures measures are in place to achieve them. It is currently in review, with a completion date of 31 December 2009.

The main areas addressed include:

- voltage fluctuations managed in accordance with Australian Standards AS/NZS 61000.3.3:1998, 61000.3.5:1998 and 61000.3.7:2001
- switching transients (voltage waveform distortion) limited where possible to less than two times normal supply voltage
- frequency variation and Country Energy's role in notifying AEMO of any sustained fluctuations
- voltage dips managed through best practice network improvement and augmentation
- steady state voltage differences between neutral and earth limited to less than 10 volts at the customer's point of supply
- lightning strikes limited in their impact on supply where possible by adherence to industry best practice system design and maintenance principles
- limitation of 'step and touch' voltage differentials managed in accordance with industry standards, namely *Electricity Association of NSW publication EC5 'Guide to Protective Earthing'* and Australian Standard HB C(b) 1
- limiting of voltage imbalance to a 6 per cent difference between the highest and lowest phase to neutral or phase to phase steady state voltages (This may be exceeded on occasion in rural areas)
- direct current generation managed in accordance with Australian Standard AS 3100
- harmonic content of voltage and current waveforms managed in accordance with Australian Standards AS 61000.3.2:1998 and AS 61000.3.6:2001, and
- mains signalling reliability set at a target of 99.5 per cent failsafe to ensure correct switching and metering functions.

A copy of *CEK8026 Electricity Supply Standard* can be downloaded from www.countryenergy.com.au.

Adoption of Australian Standard AS 60038 – 2000 Standard Voltages

CEK8026 also outlines Country Energy's adoption of the Australian Standard AS 60038 – 2000 *Standard Voltages*. The standard formally calls for a voltage range of 230V/400V plus 10 per cent, minus 6 per cent, which Country Energy will satisfy by maintaining a minimum variance of minus 2 per cent and an upper bound of plus 10 per cent.

4. Asset Management

4.3 QUALITY OF SUPPLY

4.3.1 OVERVIEW

Country Energy monitors its network by various methods including, but not limited to, use of both fixed and portable power quality loggers, electronic tariff meters, sentry devices, maximum demand indicators and portable test equipment. The network is monitored by its system operations department 24 hours a day through the use of various remote system technologies installed in zone substations, field reclosers, meters and various other devices, including sentry units located in selected premises.

We conduct site specific investigations and monitoring in response to customer initiated quality of supply complaints and enquiries.

Our specialist Power Quality Technicians, based at some 24 locations throughout regional NSW, conduct site specific investigations and monitoring in response to customer initiated quality of supply complaints and enquiries. They also work with regional network planning staff to proactively monitor network capacity and performance to deliver quality of supply.

4.3.2 PERFORMANCE DATA

Country Energy has actively participated in the Long Term National Power Quality Survey (LTNPQS), a national power quality survey conducted by the University of Wollongong and a number of other distributors throughout Australia. This survey studies parameters such as steady state voltage, voltage THD and 5th harmonics, voltage sags and voltage unbalance on three phase sites.

Power supply monitoring is advancing, with increased use of customers' meters to capture information, and Country Energy's recently introduced VIPER project has the ability to collect network performance information from some 6,500 customer meters throughout the network.

In the near future, more and more customer meters will be used to monitor network performance on a continuous basis – particularly as intelligent networks are developed – providing a broader view of performance in real time.

4. Asset Management

4.4 DISTRIBUTION RELIABILITY

4.4.1 OVERVIEW

Country Energy conducts its reliability reporting in accordance with the *NSW Design, Reliability and Performance Licence Conditions*.

Implementation of General Electric's ENMAC suite of software has been delayed slightly and is due to be completed mid 2010. Once implemented, ENMAC will achieve an automated interface between systems, producing accurate reliability information to a distribution substation level for outage reporting. Currently outage data is only recorded to a distribution feeder segment level.

Country Energy has 304 Urban Feeders, 839 Rural Short Feeders and 234 Rural Long Feeders. Over 55 per cent of our customers are on Rural Short Feeders and the average length of these feeders is greater than 60 kilometres.

Seasonal climatic impacts on the network appear to have returned to high levels in 2008/2009, with five major event days (as per *IEEE1366-2003*) in 2008/2009 compared with two in each of the previous two years.

4.4.2 ORGANISATIONAL DETAILED PERFORMANCE 2008/2009

Reliability data sets for SAIDI and SAIFI are reported for the whole organisation and feeder categories in Table 4.1.

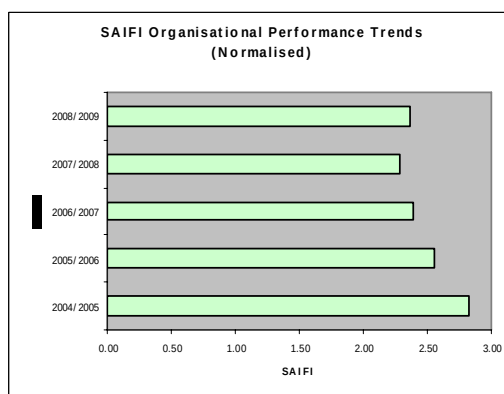
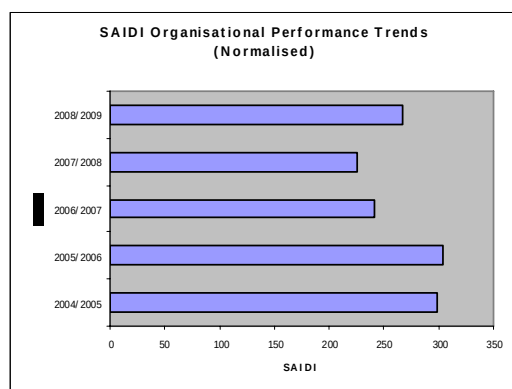
Table 4.1 – Organisational Detailed Performance 2008/2009						
Sustained Interruption Data Sets		Whole Organisation and Feeder Category				
Category		ORG	CBD	Urban	Rural Short	Rural Long
Customer Numbers		781,751	NA	220,265	433,193	128,294
SAIDI	Overall	460	NA	186	517	745
	Planned	99	NA	43	102	187
	Unplanned	343	NA	133	391	545
	Normalised	267	NA	110	285	483
SAIFI	Overall	3.17	NA	1.80	3.49	4.50
	Planned	0.39	NA	0.15	0.40	0.79
	Unplanned	2.58	NA	1.49	2.85	3.57
	Normalised	2.37	NA	1.36	2.58	3.47

4. Asset Management

4.4.3 ORGANISATIONAL PERFORMANCE (NORMALISED) TREND

Reliability data for SAIDI and SAIFI (Normalised) is reported in Table 4.2 and the graphs show the organisation trends over five years.

Table 4.2 – Organisational Performance Trends (Normalised)					
	Previous Years				Current Year
	2004/2005	2005/2006	2006/2007	2007/2008	2008/2009
SAIDI	299	304	242	225	267
SAIFI	2.82	2.55	2.39	2.28	2.37



Comment on Performance

Country Energy's network performance during the year is better than mandated in the *licence conditions* – a good result, especially considering the high level of widespread storm activity during the year, and a testament to the network improvement strategies that Country Energy is implementing.

4.4.4 FEEDER CATEGORY PERFORMANCE (NORMALISED) VERSUS TARGETS TREND

Table 4.3 – CBD Feeder Performance (Normalised) Trend						
Year		Previous				Current
		2004/2005	2005/2006	2006/2007	2007/2008	2008/2009
SAIDI	Actual	NA	NA	NA	NA	NA
	Target					
SAIFI	Actual	NA	NA	NA	NA	NA
	Target					

Table 4.4 – Urban Feeder Performance (Normalised) Trend						
Year		Previous				Current
		2004/2005	2005/2006	2006/2007	2007/2008	2008/2009
SAIDI	Actual	158	109	114	80	110
	Target		140	137	134	131
SAIFI	Actual	2.30	1.28	1.36	1.21	1.36
	Target		2.00	1.96	1.92	1.88

4. Asset Management

Table 4.5 – Rural Short Feeder Performance (Normalised) Trend						
Year		Previous				Current
		2004/2005	2005/2006	2006/2007	2007/2008	2008/2009
SAIDI	Actual	276	317	239	233	285
	Target		340	332	324	316
SAIFI	Actual	2.51	2.71	2.47	2.42	2.58
	Target		3.30	3.24	3.18	3.12

Table 4.6 – Rural Long Feeder Performance (Normalised) Trend						
Year		Previous				Current
		2004/2005	2005/2006	2006/2007	2007/2008	2008/2009
SAIDI	Actual	625	578	497	431	483
	Target		750	740	730	720
SAIFI	Actual	4.88	4.06	3.82	3.50	3.47
	Target		5.00	4.90	4.80	4.70

Comment on Performance

Country Energy is consistently better than target across all feeder categories. A return to high levels of storm activity has seen a rise in the figures when compared with the previous two years, which had quite low levels of storm activity.

Excluded Events

Table 4.7 – Excluded Interruptions for 2008/2009					
Date of Event	Description of Event	Number of Customers Interrupted	Maximum Duration of Interruption (minutes)	Effect of Event on SAIDI Figure (minutes)	Basis for Exclusion
29/12/2008	Storm Activity	27025	3508	11.48	Exceeded Tmed threshold
31/03/2009	Storm Activity	25208	4016	11.82	Exceeded Tmed threshold
20/05/2009	Storm Activity	27565	1020	8.28	Exceeded Tmed threshold
21/05/2009	Storm Activity	71474	5997	37.83	Exceeded Tmed threshold
22/05/2009	Storm Activity	21986	6190	7.61	Exceeded Tmed threshold

Major Event Day TMED

The value of TMED for 2008/2009 was 6.51 minutes.

4. Asset Management

4.4.5 INDIVIDUAL FEEDER RELIABILITY PERFORMANCE

The performance objectives for organisational average performances for each feeder category are not sufficient to identify when customers on a particular feeder experience unsatisfactory reliability performance. For this reason, SAIDI and SAIFI criteria (after 'excluded interruptions' are disregarded) act as a trigger for investigation and exception reporting purposes. The figures contained in the ministerially imposed *licence conditions* are shown in Table 4.8.

Table 4.8 – Individual Feeder Standards for Exception Reporting 2008/2009				
	Feeder Categories			
Category	CBD	Urban	Rural Short	Rural Long
SAIDI	NA	400	1000	1400
SAIFI	NA	6	8	10

Performance outside this range results in the following actions:

- immediate investigation of the causes for each feeder exceeding the individual feeder standards
- by the end of the quarter following the quarter in which the feeder first exceeded the standard, complete an investigation report identifying the causes and action required to improve the performance
- complete any operational actions identified in the investigation report by the end of the third quarter following the quarter in which the feeder first exceeded the standard, and
- where the investigation report identifies actions, other than operational actions, required to improve the performance of each feeder to the individual feeder standards, develop a project plan, including implementation timetable, and commence its implementation by the end of the second quarter following the quarter in which the feeder first exceeded the individual feeder standards.

Summarised performance against the above *licence conditions* is shown in Table 4.9.

Table 4.9 – Individual Feeder Performance Against the Standard Summary				
	Feeder Type			
	CBD	Urban	Rural Short	Rural Long
Feeders (Total Number each Type)	NA	304	839	234
Feeders that Exceeded the Standard During the Year (Total Number)	NA	15	93	38
Feeders Not Immediately Investigated (Total Number)	NA	0	0	0
Feeders Not Subject to a Completed Investigation Report by Due Date (Total Number)	NA	8	46	19
Feeders Not Having Identified Operational Actions Completed by Due Date (Total Number)	NA	1	5	6
Feeders Not Having a Project Plan Completed by Due Date (Total Number)	NA	2	17	10

4. Asset Management

Comment on Performance

Following periods of high network activity, an increased number of feeders require investigation. In all cases, the investigation commenced on time and interim remedial actions taken where appropriate, even though the report may not have been completed in the required timeframe.

The number of feeders exceeding individual feeder standards is around 11 per cent of the total number of feeders. Country Energy has implemented processes to expedite investigations. Details of each feeder not meeting the individual standards can be found in Appendix B Tables 4.11 to 4.13.

4.5 TRANSMISSION RELIABILITY

Transmission Reliability Performance Data

Country Energy has no requirement to provide this data.

5. Network Safety

5.1 OVERVIEW

The health, safety and well-being of our employees, customers, contractors, visitors, labour hire employees and the public is our highest priority. Our aim is to integrate health and safety into everything we do.

To demonstrate our commitment to health and safety we will:

- strive for an incident free workplace
- provide a safe and healthy working environment through the effective implementation of the Health and Safety Policy
- apply a risk management approach to our activities, products and services, consistent with the health and safety risk they pose
- comply with relevant legislation, regulations, standards, codes and licences
- ensure all employees are trained and have the knowledge and skills they need to undertake their work in a healthy and safe manner
- require supervisors, employees, contractors, labour hire employees and visitors to abide by all health and safety policies, procedures and other requirements
- engage in effective consultation and open communication about health and safety with our employees, contractors and labour hire employees
- conduct incident investigations fairly, with a focus on implementing preventative actions
- continually measure and improve our health and safety objectives and targets, and
- promote a 'Safety First' culture where everyone watches out for their workmates, families and local communities.

Performance Highlights

Major progress has been made in the areas of Health, Safety and Environment during the past year within Country Energy. In particular, we achieved our lowest Lost Time Injury Frequency Rate ever of 1.9 in June 2009.

Other achievements over the past year include:

- attaining continued certification to Australian Standard AS/NZS 4801:2001 *Occupational Health and Safety Management System*
- completing 1327 safety and environmental process audits focusing on premises, work teams and individuals and achieving an average score of 99 per cent against a target of 96 per cent.
- achieving reduction in the severity of employee injuries and an improvement in early intervention and management.
- implementing ongoing follow up initiatives for the *drive to zero harm* project through the Safety Alliance
- developing and delivering comprehensive regional safety plans across the business
- improving employee attendance at, and contribution to, Country Energy training sessions such as our corporate induction, practising lineworker, regulatory assessment days, and 'skilling up' sessions

5. Network Safety

- raising awareness of manual handling issues throughout the business through our Manual Handling Solutions Program
- continuing to deliver the Powerful Minds program
- gaining increasing employee involvement in health promotion activities through the Powerful Health Field Days and Pit Stop Health program pilots
- achieving ongoing success with our Powerful Apprentice Injury Prevention program
- developing comprehensive public safety fact sheets, now available on our website
- successfully integrating the TotalSAFE incident management system in May 2009
- successfully reaching five years or more without a Lost Time Injury across 85 sites, compared to 77 sites last year.

Safety for our People – Programs

Occupational Health and Safety Management System

Country Energy's Occupational Health and Safety and Environmental Management System (OHSEMS) provides a continuous improvement and quality management framework for all systems and activities within the organisation.

During 2008-2009 two independent surveillance audits of our OHSEMS were conducted by NCS International Pty Ltd, resulting in our continued certification to *AS/NZS 4801:2001 Occupational Health and Safety Management Systems* and *ISO 14001:2004 guideline for an Environmental Management System*. .

Creating a Safe Work Culture

Building on our *drive to zero harm* initiative (which focuses on leadership, employees, training, policies, procedures and standards, tools and equipment and communications) and on our successful *Stepping Up* program, a *Safety Alliance* initiative – aimed at utilising employee feedback to continue to drive cultural change within the organisation – was developed during the year. As part of this initiative, a review of safety incidents across Country Energy was undertaken and identified network operating incidents as a key focus area in the next year.

A major initiative of the Safety Alliance during 2008/2009 was the identification of network operating incidents as an area of concern. The project team will conduct a comprehensive review of the associated end to end processes. Some of the areas included in the review are:

- switching Instructions - both the creating of and application in the field;
- increasing utilisation of operating procedures and manuals supplied by OSPU;
- conducting an auditing on the authorisation process for employees to operate on the network;
- investigating and implementing "Industry best practices"; and
- reviewing and implementing a "portfolio of evidence" of network operating experience prior to re authorisation.

5. Network Safety

TotalSAFE

Country Energy captures safety incident and hazard reporting in its new Occupational Health, Safety and Environment (OHS&E) support system, TotalSAFE, which went live in May 2009.

TotalSAFE supports all of Country Energy's OHS&E functions set out by our certified management systems. It enables simpler reporting and the tracking of every safety, health, environmental and network incident.

This new system ensures improved incident management by delivering practical monthly reports, visibility on incidents and their associated actions, electronic audit inspections and hazard management.

Future Enhancements

Powerful Health

Powerful Health is Country Energy's workplace health promotion program.

While its primary goal is to decrease the risk of injury and improve the overall health of our employees, it extends well beyond the basics of health education and addresses the four cornerstones of innovative health promotion:

- policy interventions – such as guidelines for corporate nutrition that specify healthy catering requirements for all events and training sessions conducted within Country Energy.
- economic interventions – providing a \$99 corporate subsidy per employee for health and fitness activities, a \$500 grant for team based sports, subsidised influenza vaccination and skin cancer screening.
- organisational interventions – implementation and expansion of a network of employees who are nominated as Healthy Helpers, and who are allowed time in their workday to provide information about good health to co-workers, and promotional material and support for employees competing in community events such as the Sydney City2Surf. As part of this program, health field days were run in 15 locations across Country Energy between March and June 2009.
- health education – regular information about health issues provided through safety points, training sessions, intranet sites and information packs.

Powerful Health Field Days

Country Energy's *Powerful Health Field Days* was a new initiative in 2009 to raise awareness of health issues and provide on the spot health advice, information and services to our regional and field employees.

The field days were conducted at 15 locations across regional NSW and provided employees with access to activities like fitness testing, massage, cooking demonstrations and influenza vaccinations. The events have been so successful that we will continue to run similar health field days during the next financial year.

Manual Handling Solutions Program

Manual handling is a workplace hazard that causes injury. Within Country Energy, the majority of Workers Compensation claims are for musculoskeletal injuries which can be attributed to manual handling. For this reason, the Manual Handling Solutions Program was launched in February 2007, with a permanent resource allocated in January 2009 and has continued since then to increase awareness of risk management issues within Country Energy.

5. Network Safety

The traditional approach to manual handling in the workplace has been to develop and implement classes in body mechanics and / or lifting techniques. Many studies have been done in previous years showing the ineffectiveness of this in reducing injury rates. Therefore the major focus of the Manual Handling Solutions Program is to be a comprehensive risk management program with a strong consultative basis, and a focus on design issues.

The program has three main components

- Field based analysis of manual handling safety incidents of a moderate or greater severity
- An improvement / suggestion system encouraging employees to suggest solutions to manual handling in their workplace.
- Extensive communication of manual handling issues to all employees within the business.

Powerful Minds Program

Country Energy is committed to both the physical and psychological safety of our employees. Our Powerful Minds Occupational Stress program, which was developed and launched during 2007/2008, has continued throughout 2008/2009.

Powerful Minds encompasses two main areas of psychological health management of employees: occupational stress risk factors in the workplace and mental disorders. Extensive analysis of Country Energy statistical data, along with focus groups discussions with main groups, was conducted to develop the program format.

All Group General Managers and General Managers attended pilot group training in 2007, with all supervisors completing training by November 2008. The program to date has received extensive positive feedback.

Mental Health First Aid training is also being provided for key groups such as return to work co-ordinators, OHS committee members, and customer service employees dealing with the public.

Powerful Apprentice Program

As part of our successful workforce sustainability program that has created almost 740 apprenticeship positions since Country Energy formed in 2001, we have developed a collaborative safety and training group program aimed at addressing apprentice injury prevention issues.

The program's effectiveness is being trialled over a four year period between 2008 and 2011, using two groups of apprentices who joined the organisation in 2008. A target group is undertaking the program in its entirety and a control group is receiving only limited exposure, and both groups are being monitored to gauge results.

Our goal is to educate our apprentices to apply safe work practices in everything they do as they learn and become qualified tradespeople and teachers of the next generation, improving overall employee safety culture in the long term.

HealthWatch

Our *HealthWatch* program is an important initiative that identifies employees who, through their work, have experienced exposure to potentially harmful substances and promotes their long term health.

The *HealthWatch* program monitors the ongoing health of these employees to ensure an occupational disease or illness does not develop. It also enables better hazardous substance controls and work practices to be identified and put in place.

5. Network Safety

It provides an eight stage program of risk management for occupational exposure to disease:

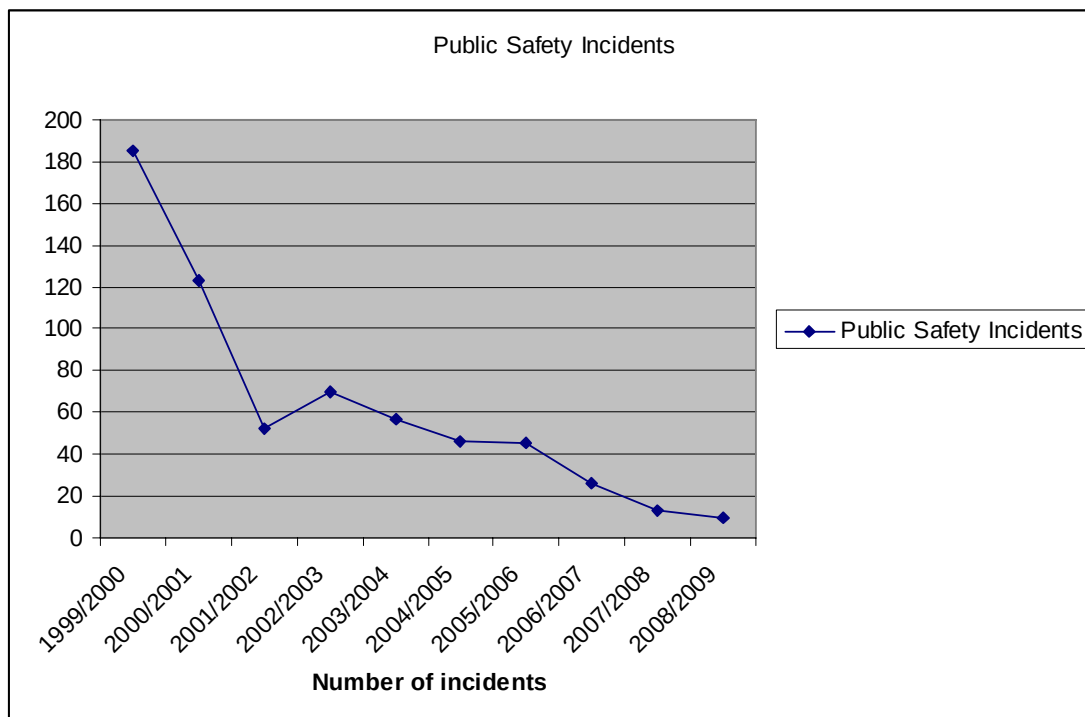
Program activity	Description
1. Recording of pre-employment medical results	A system that captures the results of all new recruits and transferees' medical results including audiometry, drug testing and general medical conditions. The current pre-placement medicals also rate an employee's risk profile which would be useful as a baseline assessment for entry point to the organisation.
2. Current medical status and register of life-threatening conditions	Conduct an organisation-wide survey to gather data on current medical status.
3. Exposure to chemicals as identified via Chemwatch and Guidelines for Health Surveillance (NOHSC 7039:1995)	Research and record potential chemical and biological hazards for high risk positions. Provide advice, education and appropriate screening interventions. Record associated hazards, screening intervals and results.
4. Vaccination status and monitoring of communicable diseases	Meet regulatory requirements. Record current vaccination status for: • Influenza Flu • Hepatitis A, B • Q Fever • Tetanus
5. Communications Plan	Meet regulatory requirements to alert employees to potential biological hazards and effects of exposure.
6. Exit Medical assessments	Maintain and record employees' health status upon exiting the organisation as insurance against potential future claims.
7. Hearing Conservation	Meet legislative and regulatory requirements by identifying hazardous noise environments and prevention of hearing damage.
8. Early Intervention Treatment	Maintain and increase access to early intervention treatment for minor work and non-work related injuries.

5. Network Safety

5.2 SERIOUS ELECTRICITY NETWORK ACCIDENTS (PUBLIC)

Country energy's Public safety program has achieved extraordinary results over its life time. The combination of targeted advertising and the direct intervention approach has been very effective. As a result the number of reportable incidents fell to a low of 10 in 2008/2009, down from 13 in 2007/2008.

This success rate makes us world leaders in the field of public safety and, with the implementation of a number of world firsts – including the overhead line flag marker – and continuing development of new approaches, Country Energy will remain at the forefront of public safety.



While we're proud of this achievement, we will continue to strive towards achieving a target of zero.

Our public safety strategy is aimed at identifying 'at risk' groups and working with them to provide targeted educational and practical assistance in managing electrical safety issues – and we provide public safety training free of charge to the main 'at risk' groups.

During the year, we worked with emergency services, earthmovers, construction workers, cotton growers, pilots, transport groups and local and State government workers. Surveys of these groups have shown 90 per cent retention of the main educational messages and information – an excellent achievement.

We've increased our Public Safety resources from four to nine to assist in continuing to target key industries like agriculture, aircraft and construction operations.

5. Network Safety

Electrical Hazard awareness sessions have been delivered to many different groups across and outside our footprint. These include

- cotton harvesting companies
- Local Councils
- NSW Rural Fire Service
- earth moving companies
- balloonists
- schools
- pilot clubs.

Electrical Hazard awareness training has also been delivered to organisations outside our footprint, as much of the activity occurs in our area.

Table 5.1 – Serious Electricity Network Accidents (Public) Trend

	Previous Years				Current Year
Category	2004/2005	2005/2006	2006/2007	2007/2008	2008/2009
Non-Fatal	2	1	2	2	1 ⁶
Fatal	1	2	2	0	0

Non- Fatal incidents

While operating a vehicle mounted crane, the crane inadvertently contacted high voltage overhead conductors, the operator was initially thrown clear. His offsider standing nearby, realising what had happened, rushed and grabbed the vehicle mounted crane controls and received an electric shock. The offsider was unable to release himself and was calling out. The original operator got up from the ground where he was thrown and grabbed his offsider but was thrown back onto the ground again. The operator got back up and used a synthetic sling to throw over the offsider and dragged him from the controls. An off duty police officer heard the calls for help and assisted the operator perform CPR on the offsider. A neighbour phoned 000 for assistance.

When the crane was fully extended with a load attached, the driver's side (northern leg) of the vehicle mounted crane stabiliser sank into the soil – they unhooked the load and relocated the stabiliser leg, lifting both the passenger side and drivers side leg off the ground and retracting the driver's side stabiliser, and placed the leg back onto firmer ground (concrete kerb) – when the leg was being extended it lifted the driver's side (northern side) up, causing the end of the crane to rise considerably. It appears that when the operator started to retract the jib of the crane, it contacted high voltage overhead conductors.

The business and people involved in this incident were based outside our footprint and, although we have completed some training outside our footprint, our ability to influence the outcomes is diminished.

⁶ Only includes incidents that are required to be reported under ENPR guidelines

5. Network Safety

5.3 ACTIONABLE ELECTRICITY NETWORK SAFETY INCIDENTS (PUBLIC)

Public safety incidents continue to decline as a result of the strategies in place to address most types of incidents. In the coming year we will be focusing on aircraft, with a series of new strategies currently being developed. Aircraft incidents form a disproportionately high number of the incidents that occur and a number of internal workshops are being held to broaden our approach.

Table 5.2 – Actionable Safety Incidents (Public) on the Electricity Network Classified by Network Element Trend					
Category of Incident	Previous Years				Current Year
	2004/2005	2005/2006	2006/2007	2007/2008	2008/2009
Overhead Mains in Position	41	35	11	9	9
Overhead Mains Fallen	2	4	4	4	
Overhead Service in Position	3	7			
Overhead Service Fallen	1		2		
Underground Mains	1	1	1		
Underground Service					
Street Light Fitting or Support					1
Cable Boxes, Pillars and Service Cabinets					
Substation (Excluding Pole-type) – Fire and/or Explosion					
Substation (Excluding Pole-type) - Other					
Substation (Pole-type) - Fire and/or Explosion					
Substation (Pole-type) - Other	1				
Poles Columns or Towers			1		
Air Break Switches					
Pole Mounted Equipment (Reclosers, Drop Out Fuses, etc)					
Service Connection or Customer Switchboard			1		
Other (Describe below)		1			
Total	49	48	26	13	10⁷

⁷ Only includes incidents that are required to be reported under ENPR guidelines

5. Network Safety

5.4 SERIOUS ELECTRICAL NETWORK ACCIDENTS (NETWORK WORKER AND ACCREDITED SERVICE PROVIDER)

A summary of performance, trends, causes and preventative action implemented are shown in Table 5.3. The network operator contestable work ring fenced operation is shown in the accredited service provider section of Table 5.3.

Table 5.3 –Serious Accidents (Network Worker and Accredited Service Provider) on the Electricity Network Trend										
	Previous Years								Current Year	
	2004/2005		2005/2006		2006/2007		2007/2008		2008/2009	
	NF	F	NF	F	NF	F	NF	F	NF	F
Network Operator Employees	5	0	6	1	1	0	4	0	1 ⁸	0
Network Operator Contractors	0	0	0	0	0	0	0	0	0	0
Accredited Service Providers	0	0	0	0	0	0	0	0	0	0

Non-Fatal Incidents

A four-man work crew was required to remove an old metal street light base from the ground and install a screw anchor in its place. A street light column was to be lifted by crane and bolted to the top of the screw anchor.

This job was being done live so that electricity supply to approximately twenty-five houses would not be interrupted.

The screw anchor was inserted into the ground and the street light column was lifted into position.

The task of screwing down one of the three mounting bolts connecting the street light column to the screw anchor was proving to be extremely difficult. The screw anchor was attempting to turn when force was being applied to the mounting bolt.

The work crew decided to insert a metal crow bar into the terminal hole of the street light column and manually hold it in position so that it would act as an opposite force to the bolt being tightened.

When the metal crow bar was placed into the terminal hole and subsequent force was being applied the end of the crow bar inadvertently pinched a live conductor inside the terminal connection area, causing the metal crow bar to become alive.

An apprentice who was holding onto the metal crow bar received an electric shock.

The apprentice appeared to be shaken and, after indicating that he was not feeling well, was taken to the local hospital for treatment. He was allowed to leave after consultation.

⁸ By definition, this should not be reported as a serious accident and it should only be reported as an actionable safety incident.

5. Network Safety

Following analysis of this incident the following improvement opportunities are recommended:

The Regional General Manager - Mid North Coast facilitates:

- reinforcement to all Mid North Coast staff the requirement to ensure that if any person receives an electric shock, no matter how minor, professional medical treatment is to be provided as soon as possible.
- availability of suitable tools (tap and die set) so that when staff encounter faulty threads on equipment they have the resources to correctly remedy the situation without having to resort to unacceptable work practices.
- review of the work practice of working live, while removing old metal street light column bases from the ground, installing screw anchors, erecting, securing the street light column and electrically connecting the street light, to ensure that the task can be carried out safely.
- if it is deemed that the work can be carried out safely, carry out a review of existing SWMS's to consider whether they are adequate for the task of replacing street light bases and columns with the electricity still connected (i.e. working live). If the review determines that existing SWMS are not adequate, facilitate the development of a suitable SWMS.

The Area Manager - Taree facilitate:

- reinforcement to all staff on site that when a risk assessment of a worksite is undertaken and it is identified that a supervisor is required for an apprentice and/or an assistant is required, that the initials of the nominated supervisor and/or assistant is written onto the HIRAC form in the space provided, and to ensure that all hazards, risks and controls are documented, on this occasion a tree had to be trimmed to allow for crane access, There is no indication on the HIRAC form assessment of a tree being trimmed.
- reinforcement to all staff on site the requirement to continually assess the task they are undertaking to ensure they identify and document all foreseeable hazards and introduce measures to eliminate or reduce the risk to the lowest level possible using the 'hierarchy of controls' as required by OHS Regulation 2001.

5.5 ACTIONABLE ELECTRICITY NETWORK SAFETY INCIDENTS (NETWORK WORKERS)

Following an analysis of the actionable electricity network safety incidents data over the previous three years, it is apparent that the majority of incidents involved overhead mains in position and were a result of a failure to adhere to established and documented work procedures.

Country Energy has recently delivered the Practising Powerline Workers Course. This course merges the technical knowledge and skills of the new apprentice powerline workers with the ageing qualified line workers, using this connection to make some fundamental changes to behaviours and challenging acceptance of poor work practices.

The overall objective is to enhance professional, technical and behavioural competencies of Country Energy's Powerline Workers through quality training/mentoring and education.

Following on from this, a new Practising Powerline Workers course is currently in development which will further upgrade the technical skills of our older work force. The continued decrease in incidents from 2007/2008 can be attributed to the success of this course along with an increased focus on our safety process auditing.

5. Network Safety

Table 5.4 – Actionable Safety Incidents (Network Workers) on the Electricity Network Classified by Network Element Trend

Category of Incident	Previous Years				Current Year
	2004/2005	2005/2006	2006/2007	2007/2008	2008/2009
Overhead Mains in Position	2	7	15	5	2
Overhead Mains Fallen		1	2		
Overhead Service in Position		2			1
Overhead Service Fallen					
Underground Mains	2	2		3	
Underground Service					
Street Light Fitting or Support		1	1		
Cable Boxes, Pillars and Service Cabinets		1			1
Substation (Excluding Pole-type) – Fire and/or Explosion	2				
Substation (Excluding Pole-type) – Other				1	
Substation (Pole-type) - Fire and/or Explosion		1			
Substation (Pole-type) – Other			1		1
Poles Columns or Towers					
Air Break Switches			2		
Pole Mounted Equipment (Reclosers, Drop Out Fuses, etc)		1	1		
Service Connection or Customer Switchboard			2		
Other (Describe below)					
Total	6	16	24	9	5

5. Network Safety

5.6 MAJOR INCIDENT REPORTS

Network operators are required under the *licence conditions* to report to the Minister for Energy within 24 hours any major network incidents involving significant injury to persons, loss of property or widespread supply interruptions. High level severity incidents are to be advised immediately.

Network operators shall list all major incident reports made to the Minister as required under these *licence conditions*.

Table 5.5: Summary of Major Incident Reports Submitted for 2008/2009

Date	Incident Description	Locality
16/07/2008	Tractor hit pole Narooma	Narooma
18/08/2008	Automatic load shedding of Country Energy's Lismore load.	Lismore
13/09/2008	Contractor excavator contacted 11 kV line	Wagga
21/09/2008	Aircraft hit 11 kV line	Paupong
20/09/2008	Circuit Breaker Failure at Kempsey TransGrid	Kempsey
21/09/2008	Widespread tripping at Grafton zone subs	Grafton
18/09/2008	Bulldozer contacted 11 kV line	Bowraville
18/08/2008	Automatic load shedding of Country Energy's Lismore load.	Lismore
16/10/2008	Widespread tripping of Port Macquarie Load	Port Macquarie
5/11/2008	Truck contacted 22 kV line	West Wyalong
10/12/2008	Harvervester contacted line	West Wyalong
12/12/2008	66 kV Bus tripped at Tamworth BSP	Tamworth
6/02/2009	Contractor crane contact 66 kV conductor	Bungendore
18/02/2009	Tripping of Boronia St zone substation	Port Macquarie
25/02/2009	11 kV Feeder trip - Kempsey - Aldavilla	Kempsey
26/02/2009	Goulburn - 33 kV in to LV	Goulburn
4/03/2009	Nambucca - LOS to zone substation due to lightning	Nambucca
4/03/2009	Whitbread St- LOS to zone substation due to lightning	Taree
20/03/2009	LOS to Zone Subs Taree area SENI-2 Emailed	Taree
30/03/2009	LOS to Raleigh, Sawtell	Coffs harbour
1/04/2009	Multiple outages to customers Mid and Far Nth coasts	Coffs harbour
31/03/2009	Car vs pole in Taree area	Taree
31/03/2009	Raleigh - storms - customers off 12 hours plus	Raleigh
1/04/2009	Grafton - storms, floods - customers off 12 hours plus	Grafton
1/04/2009	Braidwood - SENI-2 'S sent. Floods 12 hours plus	Braidwood
1/04/2009	Karangí - SENI-2 'S sent. Floods 12 hours plus	Karangí
1/04/2009	Multiple outages to customers Mid and Far Nth coasts	MN Coast
4/04/2009	66 kV Bus outage at Ballina zone sub	Ballina
20, 21 & 29/04/2009	Loss of supply to Raleigh, Sawtell (2 outages, seni-2's)	Coffs harbour
28/06/2009	Loss of supply to Wagga Substations	Wagga
14/06/2009	Bus accident affecting 2 Ewingsdale feeders	Byron bay
20/05/2009	Storms NSW North coast	NSW North
13/07/2008	Employee received shock when removing LV link assembly	Ballina
3/11/2008	Pole failed and fell, lineworker suspended in harness	Broken Hill
23/02/2009	Employee received flash burn when drill contacted cable	Inverell
3/03/2009	Crane operator made contact with LV street conductor	Wingham
19/03/2009	Employee received shock from capacitor bank	Queanbeyan

6. Customer Installations

Monitoring of installations is carried out in accordance with regulations and industry codes of practice. In addition to the *Electricity Supply (Safety and Network Management) Regulation 2002*, installations must be consistent with the *Electrical Safety (Electrical Installations) Regulation 1998*, the *Code of Practice for Installation Safety Management*, and the *Code of Practice for Contestable Works Accreditation*.

Country Energy's *CEPG8004 Customer Installation Safety Plan* outlines the approach taken to managing all aspects of customer installation work. The plan defines our approach with respect to the roles of both contractors and Country Energy.

Installation and inspection data is managed within Country Energy's Contractor Performance Register which enables management of NOEW and NOSW information and the auditing of individual contractors to ensure adherence to *CEPG8004*.

6.1 REPORTS AGAINST CUSTOMER INSTALLATION SAFETY PLANS

Table 6.1 - Installation Inspections Trend					
	Previous Years				Current Year
	2004/2005	2005/2006	2006/2007	2007/2008	2008/2009
Number of Notifications (NOEW)	27,240	23,387	20,651	19,721	18,771
Number of Inspections	11,644	11,221	9,404	8,555	7,297
Installation Inspection Rate (%)	48.7	48.0	45.5	43.4	38.9%
Major Safety Defect Rate (%)	2.0	3.0	3.3	2.4	2.8%
Safety Breach Notices Issued (%)	0.7	11.8	3.4	2.5	2.9%
Number of Warnings Issued	235	306	323	212	214
Reports to Office of Fair Trading (No.)	9	7	5	5	4
Number of Audits by Network Operator	102	354	68	37	28

Note: Number of Notifications (NOEW) now applies to Certificates of Compliance Electrical Work (CCEW)

6.2 CUSTOMER INSTALLATION SHOCK REPORTS

Data recorded for this period indicates a 14 per cent decline in the total number of reported public shocks from the previous period. A project considering the replacement of targeted overhead services is expected to reverse the increased trend in the "Faulty Service Joint" category. A small increase trend relating to the categories involving human interaction can be seen from the data. Storm/Lightning category has had an increased contribution related to a number of severe storm events in this period in the Mid and Far North Coastal regions. Additional cause categories would improve the expansion of the "Other" category for both Network and Installation related incidents.

Table 6.2 - Customer Installation Shock Reports Trend					
	Previous Years				Current Year
	2004/2005	2005/2006	2006/2007	2007/2008	2008/2009
Shocks on Customer's Premises (Number Reported)	606	696	608	834	719

6. Customer Installations

Table 6.3 Customer Installation Safety – Categories of Shocks Analysed for 2008/2009

Category	Number		% of Total
	Fatal	Non-Fatal	
Contact with Consumer's Mains – Faulty Mains		4	0.6
Contact with Consumer's Mains – Human error		13	1.8
Contact with Live Parts at Switchboard – Faulty Switchboard		0	0
Contact with Live Parts at Switchboard – Human Error		17	2.4
Unsafe Installation Work by Licensed Contractor		21	2.9
Failure of Part of Installation (not water related)		32	4.4
Defective or Unsuitable Appliance		39	5.4
Working on or Interference with Installation		4	0.6
Working on or Interference with Appliance		1	.1
Water Damage or Ingress		16	2.2
No Cause Found (Including Static Electricity)		81	11.3
Other (described in Table 6.3a)		491	68.3
Total		719	
Total per 1,000 customers		0.9	

Table 6.3a - Categories of 'Other' Shocks Analysed for 2008/2009

Category	Number		% of Total
	Fatal	Non-Fatal	
Cause Category Installation Related			
Faulty Mains Box		56	7.8
Poor Earthing		24	3.3
Other		15	2.1
Subtotal		95	
Cause Category Network Related			
Faulty Mains Joint		62	8.6
Faulty Service Joint		132	18.4
Faulty Underground Joint		20	2.8
Long LV Run		25	3.5
HV Leakage		4	0.6
Incorrect Polarity		2	0.3
Other		25	3.5
Sub total		270	
Other Cause Categories			
Lightning/Storm		29	4
Nuisance Tingles <10 volts		70	9.9
N/A		26	3.6
Subtotal		126	
Total		491	

7. Contestable Works Scheme

Country Energy continues to use the Contractor Performance Register to monitor all Level 2 works. Level 1 and Level 3 work is collected regionally by the Quality Control Co-ordinators, working in conjunction with the regional Planning and Customer Connection Managers.

2008/2009 has seen the following trends:

- **Level 1** – Internal project notifications are down and inspection rates falling in line with this decline. This is partly explained by the increase in external Level 1 works notices and inspections.
- **Level 2** – The 2008/2009 trends in project notices, inspections and non-conformances are similar to previous years.
- **Level 3** – A significant increase in certified external designs is reflective of the increase in Level 1 external works. As more external Accredited Service Providers continue to gain accreditation to operate within the service area this increase will continue to meet demand of contestable market.

Table 7.1 – Contestable Works Trend										
	Previous Years								Current Year	
	2004/2005		2005/2006		2006/2007		2007/2008		2008/2009	
	Int	Ext	Int	Ext	Int	Ext	Int	Ext	Int	Ext
Network Work (Level 1)										
Project Notifications	2,950	228	2,336	396	1,861	799	1,520	1,103	778	1,269
Initial Inspections of Completed Projects	1,565	195	1,729	414	1,048	618	540	623	786	947
Of Projects Inspected, Number Initially Nonconforming	431	31	389	45	175	85	109	146	220	200
Customer Connection Work (Level 2)										
Notifications (NOSW)	2,300	21,193	6,780	13,028	6,299	13,203	5,573	13,238	5,518	17,135
Inspections by Network Operator	1,514	1,992	4,321	9,456	3,540	9,268	3,143	11,807	3,485	12,919
Major Defects	1	42	6	117	3	130	0	102	3	107
Network Design Work (Level 3)										
Designs Certified	2,950	253	2,336	410	1,861	631	1,225	1,114	778	947

8. Bushfire Risk Management

Overview

Country Energy is committed to achieving effective asset management by implementing strategies to ensure the safe and reliable operation of the network. A key component is to manage our assets in a manner which minimises the risk of bushfires, protects assets and maintains the highest possible supply reliability during bushfire risk periods.

The objectives of Country Energy's Bushfire Risk Management Plan are to:

- ensure public safety
- establish standards for vegetation management near electricity lines, particularly in bushfire prone areas
- reduce electricity supply interruptions related to vegetation
- minimise the possibility of fire ignition due to electricity lines and associated equipment
- minimise any other adverse impact of power lines on vegetation, and
- ensure balance between bushfire risk management, vegetation management and the provision of a safe, reliable and adequate supply of electricity of appropriate quality.

The Bushfire Risk Management Plan covers the following key issues:

- identification of bushfire prone areas and network assets capable of initiating bushfires managed by a system which ensures this information is kept up to date
- asset maintenance procedures
- vegetation management strategies and procedures
- equipment and construction standards as related to bushfire mitigation
- information relating to the rights and duties of customers who own private powerlines and the dangers of trees coming into contact with those lines
- asset maintenance procedures for private powerlines and provisions to ensure maintenance standards are enforced by the network operator (Country Energy in this case)
- management of complaints in relation to bushfire risk management and procedures to ensure appropriate investigations and remedial actions are undertaken as required
- liaison and consultation with other organisations in relation to bushfire risk management issues
- information for the general public about fire hazards associated with overhead powerlines and vegetation
- special operating procedures and precautions during times of high fire danger, and
- reporting to the NSW Minister for Water and Energy.

Digital bushfire prone area mapping data for NSW is available in Country Energy's geographic information system. This information facilitates better operational management during fire ban periods and is also used to target preventative programs, such as the fitting of low voltage spreaders.

8. Bushfire Risk Management

Asset Inspection

Country Energy's asset inspection and assessment process includes:

- inspection and assessment of overhead poles and structures
- visual inspection of overhead powerlines, including an assessment of all associated assets such as cross arms, conductors, pole mounted equipment and the above ground components of underground systems
- treatment of poles
- earthing integrity checks, and
- identification of vegetation encroachment issues.

Private assets are recorded in Country Energy's asset management system and inspected on the same cycle as other assets. At 30 June 2009 the total pole population was 1,373,649 with 77,115 of those being privately owned.

The results of the 2008/2009 asset inspection program are shown below

Table 8.1 – Asset Inspection Program 2008/2009		
Poles Inspected	Poles Replaced	Poles Reinforced
357,545	6,123	931

This represents an average inspection cycle of 3.7 years, which complies with Country Energy's policy of a 4.5 year average cycle. The average pole replacement/reinforcement rate of 0.51 per cent overall or 1.97 per cent of poles inspected is consistent with figures for previous years.

Annual Bushfire Patrol

Annual bushfire audits are carried out prior to the normal commencement of the bushfire season using a variety of methods including fixed wing aircraft, helicopter and ground based patrols. All system defects observed, including those that are not likely to have an associated fire risk, are recorded in the asset management database with an appropriate severity rating indicating the level of risk and the required repair time. 'Urgent risk' defects require rectification within fourteen days. 'Risk' defects do not carry an immediate concern for safety or reliability but are scheduled for rectification within six months.

The results of the annual pre-summer bushfire patrols for the 2008/2009 season are shown below.

Table 8.2 – Annual Patrol Defects Identified 2008/2009			
Defect Category	Urgent Risk	Risk ⁹	Total
Number of defects identified in annual patrols	349	18,962	19,311
Defects rectified by 30/06/09	349	4,050	4,399

In relation to bushfire risk issues Country Energy does not differentiate between distributor and privately owned Network reticulation assets. Owing to the legal issues and associated time delays that can be involved in the process of forcing customers to rectify defective installations, it is considered to be more effective in terms of risk avoidance for Country Energy to attend to private asset defects in conjunction with the remedial work on our own distribution system.

⁹ The number of "Risk" defects reported here are defects identified as not likely to have an associated fire risk. These risk defects are scheduled for rectification within 6 months of it being reported.

8. Bushfire Risk Management

Fire Reports

All fires attended by Country Energy employees are recorded in a network incidents database, including those in which Country Energy assets have been damaged and/or the alleged cause of ignition involved Country Energy or private line assets.

Table 8.3 – Fire Ignitions Attended by Country Energy Personnel	
Arson	3
Property owner burning off crops	3
Electrical fault	30
Equipment failure or malfunction	41
Pole top fire	7
Vegetation	0
Vehicle impact	7
Customer installation	18
Undetermined	56
Total	165

Notes:

- Electrical faults include fault current, bird strike and blown debris
- Pole top fires include lightning strikes and leakage
- Vegetation includes trees and limbs falling into the clearing space
- Vehicle impact includes farm machinery, high loads and aircraft
- Customer installation includes disconnection for house fires

Low Voltage Spreader Program

Following the addition of the bushfire prone area mapping data to Country Energy's geographic information system in early 2006 queries were run to identify multiple phase, bare conductor and low voltage spans in excess of 45 metres in fire prone areas. Over 6,000 spans were fitted with low voltage spreaders under this initiative.

In areas where data was incomplete low voltage data capture programs have been implemented and asset inspection employees are generating work tasks for the installation of low voltage spreaders on susceptible spans during the course of the normal asset inspection cycle.

8. Bushfire Risk Management

Community Awareness

Specific awareness campaigns which encourage reporting of vegetation in proximity to power lines by the public are run by Country Energy immediately prior to, and during, the bushfire season.

A general safety media schedule runs throughout the year and targets seasonal activities through print and electronic media, and in specialist industry publications to reach key target groups such as heavy plant operators.

Bushfire related electrical safety topics included:

- bushfire and storm safety tips
- dangers of trees near power lines
- electrical safety for emergency services personnel
- advice on safety when burning off
- safe operation of farm equipment such as grain augers, harvesters and irrigation systems, and
- advice on electrical safety for heavy plant operators.

Country Energy's general education programs include:

- community consultation through on-site meetings with Country Energy qualified staff to address specific vegetation issues
- tree planting guidelines
- advice on vegetation management
- liaison with stakeholders including property owners, government departments, bushfire associations and interested community groups
- attendance at community meetings
- presentations at field days and workshops, and
- bushfire safety information on Country Energy's website.

Our Public Safety team also deliver sessions to the public – in particular, groups directly at risk of inadvertently interfering with the network – and provides them with positive strategies to assist them in avoiding incidents that may initiate bushfires.

During the 2008/2009 year the public safety team delivered direct intervention type training to 3,342 people across our footprint.

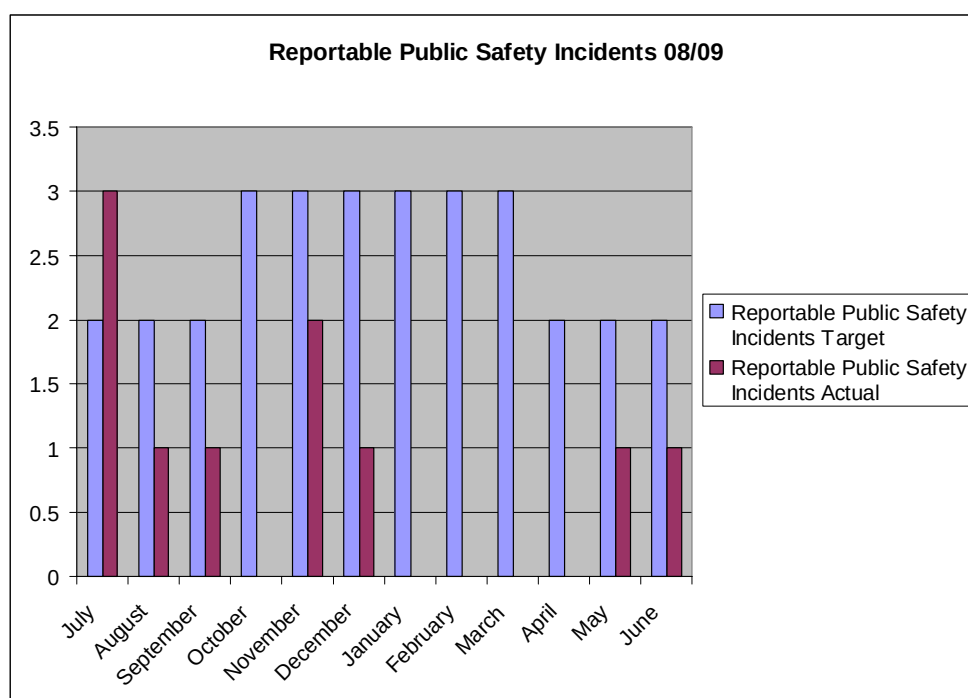
9. Public Electrical Safety Awareness Campaign Reports

9.1 PUBLIC SAFETY - IMPROVED COMMUNITY SAFETY PERFORMANCE

The number of reportable incidents fell to a low of 10 in 2008/2009, down from 13 in 2007/2008. Although these statistics show a continued decrease in incidents it is necessary to eliminate all incidents to meet our corporate strategies and to satisfy community expectations.

The strategy of identifying the 'at risk' groups and then educating and assisting those groups to solve their electrical safety issues is a key contributing factor in the decline of incidents. These 'at risk' groups include emergency services earthmovers, construction workers, cotton growers, pilots, transport groups, local and state government workers and the like. Regional Safety, Environmental Audit & Investigation Coordinators (SEALCs) are now responsible for Public Safety within their respective regions. This has effectively increased Public Safety resources from four to nine.

Key remaining areas of incidents continue to be agriculture, aircraft and construction operations.



Four years ago the Public Safety Team implemented a follow up survey of a small random sample of the participants to assess training effectiveness. This survey is based on participants remembering the key points delivered during the awareness sessions after a three month period.

Retention of the key learning outcomes continues to be outstanding, at around 90 per cent over the last year. To help with retention, presenters utilise a power point presentation, practicable demonstrations and a reinforcing DVD exposing the participants to the key points in three different ways.

A recall rate of 15- 20 per cent is considered to be normal after three months. This program structure delivers the learning in multiple forms and also delivers an understanding so that the ability to recall is less important in knowing how to respond.

During the 2008/2009 year the public safety team delivered direct intervention type training to 3,342 people across our footprint.

9. Public Electrical Safety Awareness Campaign Reports

The training by the public safety staff is provided free of charge to the target organisation. After consultation with the organisation it is presented in a practical and relevant manner targeted directly at the risk group.

During the year the Public Safety Team:

- provided direct face to face electrical safety awareness sessions to over 1,500 people in 'at risk' groups
- continued distribution of '3,6,8 clearance' stickers for operators in accordance with the WorkCover *Code of Practice - Working near overhead power lines*
- updated Public Safety fact sheets on the Country Energy's website giving easy access to important safety information - <http://www.countryenergy.com.au/> - Switch onto Safety: Community, Emergency Services, Agri-Business, Construction, Vegetation Management and Natural Gas Safety
- updated a static display showing the hidden hazard caused by machinery coming in contact with overhead high voltage powerlines
- provided safety information packs to individuals and groups involved in line strike incidents
- continued utilisation of a scaled power line model to demonstrate indications of possible hazards when attending an incident where electricity may be involved
- attended eleven industry and agricultural field days as well as attending a host of local shows/ field days
- actively promoted the WorkCover *Guide to Working near overhead power lines* across our region
- provided electrical hazard awareness training to NSW Rural Fire Service, Newcastle Light Aircraft Club , RAA Balloonists and Cane Harvesting groups, and
- participated in 'at risk' group industry working groups to imbed electrical safety into their standard work practices.

Driving the number of incidents lower over the next year will present a challenge with a significant increase in agricultural activity expected with continuing rains.

By the process of direct intervention, the cornerstone of the Country Energy Public Safety program, we have achieved a remarkable decline in Public Safety incidents.

The communications strategy necessary to achieve these outcomes is both varied and tailored to the specific 'at risk' group and organisation in order that the intervention is successful and to ensure safe work practices are adopted.

The Country Energy Public Safety Plan is reviewed and updated annually and is an accurate picture of the very effective incident reduction strategies we have in place.

10. Power Line Crossing of Navigable Waterways

10.1 EXISTING CROSSING NUMBERS

Crossing numbers represented in Table 10.1 are presently being revised in accordance with the NSW Maritime GIS navigable overlay.

A list of the 2,608 crossings that have been identified by Maritime as being navigable are presently being analysed by Network Assets to identify the numbers of overhead and submarine crossings in the respective field service centres. Once complete, the lists will be distributed to the Field Service Centres for a field check.

Each crossing will be looked at and validated as to if it is a 'Navigable Crossing' or not and documented as such using the field check list that will be provided. In some cases when an opportunity arises and to speed up the process aerial photos will be taken rather than using land travel for the 'first pass'.

Once inspected a desktop assessment will be carried out on each of the crossings. The crossings considered by Country Energy as being 'Non Navigable' will be listed. The 'Non Navigable' list will be tabled with NSW Maritime to review and confirm the 'final' status of the crossing.

The program to augment or relocate the navigable crossings will be prioritised according to the risk assessment results.

Table 10.1 – Power Line Crossings of Navigable Waterways Summary					
	Existing (Number)	New (Number)	Incidents (Number)	Crossings Reconstructed (Number)	Crossings Identified as Requiring Conversion to Submarine Crossings (Number)
Overhead Crossings	2,470		2	3	2
Submarine Crossings	138				

Note: The existing numbers of overhead and submarine crossings shown in Table 10.1 are based on figures provided by the Network Asset Group using the NSW Maritime GIS overlay.

Proposed Planning Works:

- All new work to be in accordance with the new NSW Maritime Code and Country Energy's policy.
- Update Country Energy Procedural Guideline for Crossing of NSW Navigable Waters CEPG2013 in accordance with both the new Maritime codes and Country Energy inspection criteria.
- Update Country Energy' Subtransmission and Distribution Planning Manuals to reflect the new codes and actions required.
- Verify the numbers of actual navigable overhead and underground crossings within the Country Energy network. NSW Maritime are prepared to assist once it is established what crossings need reassessment
- Prioritise existing crossings and carry out risk assessments according to the NSW Maritime Code. This will include an inspection of vegetation, the confirmation of existing data, inspection of supporting structures and hardware, conductor condition, height verification and signage requirements. Special attention will be given to the hardware supporting conductors in coastal areas. This will include the inspection of all concealed supports (bolts through poles, crossarms etc).
- Look at new design criteria that will assist in preventing corrosion in extreme coastal environments.
- Carry out risk assessments on all crossings within the Country Energy network and alterations to the crossings as necessary.

10. Power Line Crossing of Navigable Waterways

- Arrange for signage to be manufactured to the NSW Maritime standards (less the maximum vessel height figure) and stored in an appropriate place. The signs will be available for designers as required. Crossing signs should include the two types to be placed under the crossing, the two types for flood areas, the two types for submarine crossings and the advisory launching site signage. The maximum vessel height for the crossing will be marked on the under crossing signs by an approved method prior to installation.
- Update all in house data to reflect what is in the field. Every crossing in the database should have all data relevant to the crossing including designs, photos, hardware used, the risk assessment conclusion and any history relevant to the crossing.
- Ensure Country Energy employees are aware of the new codes and reporting protocols that apply with all new crossing work, changes that occur and any incidents that might occur on the network.
- Ensure external design employees consulting for Country Energy are designing crossings in accordance with the NSW Maritime Code.

10.2 CROSSINGS INCIDENTS

Country Energy has two recorded incidents for 2008/2009:

1. Crossing Incident Brushgrove:

a) The Incident:

At 1300 hours on 22 May 2009 and during a major flood on the Clarence River, a yacht dragged anchor and the mast impacted with the 11 kV three phase crossing on the South Arm of the Clarence River next to the Wingfield Bridge. The feeder protection operated correctly and isolated the line. The yacht cleared the crossing and held anchor between the crossing and the nearby Wingfield Bridge.

b) Actions:

- the crossing conductors were damaged and replaced a few days later when the river crossing became accessible.
- a risk assessment of the crossing was carried out and it was decided that the crossing posed an 'intolerable risk' due to the height of the crossing (even though higher than the bridge) and the numbers and frequency of yachts that frequently anchor overnight in the area.
- Country Energy's planners and designers were notified.
- the crossing will be relocated as soon as possible, likely with a cable across the nearby Wingfield Bridge.

2. Crossing Incident South West Rocks:

a) The Incident:

At 1205 hours 31 January 2009 a yacht proceeding up the Macleay River at South West Rocks contacted the 11 kV three phase overhead line supplying the Stuarts Point area. The feeder protection operated correctly and isolated the line. Country Energy attended the site to assess the line and re-establish supply.

b) Actions

- an onsite check was carried out on the crossing and a height of the lowest conductor was measured at 16 metres.
- the mast height of the vessel was stated as being 22 metres and the crossing signs indicated 11.5 metres.

10. Power Line Crossing of Navigable Waterways

- a risk assessment was carried out on the crossing indicating the crossing posed an 'intolerable risk' because of its proximity to the ocean and waterway use and therefore should be relocated.
- Country Energy's planners and designers were notified.
- as an interim measure and until the crossing can be removed, new signage manufactured to the new NSW Maritime Code will be installed and appropriate conductor markers positioned.
- the minimum estimated time for relocation is 12 months.

Appendix A

Country Energy

ELECTRICITY NETWORK PERFORMANCE REPORT 2008/2009

Declaration by Managing Director

In submitting this Electricity Network Performance Report (the Report), I declare the Report:

1. Complies with reporting requirements prescribed under the *Electricity Supply (Safety and Network Management) Regulation 2008*, and the "Distribution Network Service Provider Annual Report Outline" (the Outline), as provided by the Industry and Investment NSW.
2. Has been appraised by an independent appraiser in accordance with the "Appraisal Guidelines" contained in the "Network Operator Report Outline"
3. Includes a copy of the appraisal, signed by the independent appraiser
4. Has been checked in accordance with recognised quality procedures; and in my opinion, there are reasonable grounds to believe the data, and notes in respect of data contained in this Report, give a true and fair view of the organisation's performance in respect of the matters contained in the "Network Operator Report Outline".

SIGNATURE:



Craig Murray
Managing Director

DATE: 26 NOVEMBER 2009

Appendix B

Appendix B - Table 4.11: Urban Feeders Which Exceeded the Individual Feeder Standards During the Previous 12 Month Period

SAIDI Standard: 400; SAIFI Standard: 6

Feeder Name	Location	Customer Numbers	Length	Date of First Non-Compliance	SAIDI		SAIFI		Description of Non-Compliance and Reason	Remedial Actions Proposed or Taken Including Timetable
					ANC	Current	ANC	Current		
BBYG2	BBYG2 Batehaven	1606	10.0	30/06/2008	441	256	4.05	2.34	Fauna and third party.	Line inspection. Review of protection settings. (Complete) Installation of a recloser.
CHN3B8	CHN3B8 Nth Jetty	637	6.9	30/09/2008	443	364	3.82	2.14	Burnt off 11kV bridging wire.	Vegetation Management. Carry out thermoscan of feeder when loaded (Completion December 2009).
DEN32	DEN32 Industrial	345	22.2	31/12/2008	407	358	2.19	2.10	Equipment failure on Subtransmission Line.	Pole & line defect management. Vegetation management. Refurbishment of 66kV incoming feeder to Deniliquin Z/S. Refurbishment works on Deniliquin Urban 22kV feeders.
EWE3B6	EWE3B6 Byron No.1	1472	12.31	30/06/2009	444	444	5.01	5.01	Equipment failure, fauna and third party contact.	Maintain Normal Inspection & Maintenance, Vegetation Maintenance Cycle - 1 yrs. Construction of feeder tie to Lawson St Byron Bay. (Completion June 2010)
GNF32	GNF32 Grefell Town No.2	342	3.1	31/03/2009	534	534	3.12	3.11	Storms and lightning caused pole failure on subtransmission feeder.	Investigation report to be finalised.
GTH22	GTH22 No.2 Yenda	5	15.3	31/03/2009	827	827	2.00	2.00	HV conductor failure in lightning storm and fauna.	Refurbishment of Yenda 33kV line. Construction of alternate 33kV supply to Yenda. (Completion November 2009)

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					ANC	Current	ANC	Current		
MWB292	MWB292 M29 Moomin Cotton Gin	2	1.6	30/06/2008	475	289	1.50	4.00	Fauna.	Line Maintenance. (Complete)
STL3B5	STL3B5 Baylon	954	10.4	30/06/2009	436	436	6.66	6.66	Storms. Subtransmission line outages while network in abnormal configuration.	Construction of New Boambee 132/66/11kV Zone Substation. (Completion November 2009)
STL3B7	STL3B7 Sawtell Town	2148	11.6	30/06/2009	729	729	5.63	5.63	Crossarm failure & storms. Subtransmission line outages while network in abnormal configuration.	Construction of New Boambee 132/66/11kV Zone Substation. (Completion November 2009)
STL3B8	STL3B8 Industrial	75	0.3	30/06/2009	433	433	5.00	5.00	Storms. Subtransmission line outages while network in abnormal configuration.	Vegetation management & aerial patrol. Construction of New Boambee 132/66/11kV Zone Substation (Completion November 2009).
TSH3B5	TSH3B5 West Tweed	2100	10.8	31/03/2009	539	589	3.00	4.04	Conductor and connection failures and fauna.	Maintain Normal Inspection & Maintenance, Vegetation Maintenance Cycle - 1 yrs. Replacement of 1.2km of conductor. (Completion June 2010)

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					ANC	Current	ANC	Current		
WTR2Z09	WTR2Z09 Gins	2	0.4	31/03/2007	569	147	5.50	3.00	Storms.	Performance to be monitored.
YAM3B3	YAM3B3 Yamba Sth	884	5.3	30/09/2008	497	560	3.04	4.04	11KV cable failure.	Maintain Normal Inspection & Maintenance, Vegetation Maintenance Cycle - 2 yrs. Construction of an underground tie line. (Completion December 2009)
YEN3B5	YEN3B5 Yenda Town	371	4.9	31/03/2009	831	861	2.16	3.01	HV conductor failure in lightning storm and fauna.	Construction of new Subtransmission line to Yenda Z/S. (Completion November 2009)
YOU3B4	YOU3B4 Young Town 7	1128	11.0	31/03/2009	536	522	5.17	4.17	Storms / lightning caused equipment failures.	Defect rectification on the subtransmission feeder. Zone substation refurbishment (Complete). Extensive reconductoring.

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Appendix B - Table 4.12: Rural Short Feeders Which Exceeded the Individual Feeder Standards During the Previous 12 Month Period

SAIDI Standard: 1000; SAIFI Standard: 8

Feeder Name	Location	Customer Numbers	Length	Date of First Non-Compliance	SAIDI		SAIFI		Description of Non-Compliance and Reason	Remedial Actions Proposed or Taken Including Timetable
					ANC	Current	ANC	Current		
AHY2T05	AHY2T05 Gins	133	46.8	31/03/2009	1229	1229	6.76	6.76	Storms and lightning.	Defect rectification and vegetation management.
AVE3B2	AVE3B2 Alstonville /Tuckombil	1679	51.5	31/03/2009	1261	1382	3.38	4.34	Storms / lightning and fauna.	Maintain Normal Inspection & Maintenance, Vegetation Maintenance Cycle - 3 yrs. Installation of a recloser and replacement of conductor at various locations. (Completion June 2010)
AVE3B5	AVE3B5 Alstonville Nth	1501	48.3	31/03/2009	1095	1124	8.81	8.73	Storms / lightning and fauna.	Maintain Normal Inspection & Maintenance, Vegetation Maintenance Cycle - 3 yrs. Installation of a recloser and replacement of conductor at various locations. (Completion June 2010)
BAR12	BAR12 East Barham	108	39.4	31/03/2009	1315	1315	3.97	3.97	Storms / lightning caused equipment failure and zone substation battery charger failure.	Refurbish Barham 66kV subtransmission line. (Completion December 2010)
BAR32	BAR32 Gonn	221	131.4	31/03/2009	1229	1230	3.06	3.06	Storms / lightning caused equipment failure and zone substation battery charger failure.	Refurbish Barham 66kV subtransmission line. (Completion December 2010)
BAR42	BAR42 Barham	772	21.9	31/03/2009	1212	1211	3.01	3.01	Storms / lightning caused equipment failure and zone substation battery charger failure.	Refurbish Barham 66kV subtransmission line. (Completion December 2010)

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					ANC	Current	ANC	Current		
BBA656	BBA656 Gladstone	88	142.8	30/09/2008	1063	780	4.44	3.05	Strong winds, storms and lightning.	Installation of 2 reclosers.(Completion December 2011)
BBI1101	BBI1101 M11 Ghoolend aadi	42	98.1	31/03/2009	2351	2621	8.33	9.24	Storms and lightning.	Pole replacements and defect rectification (Completion October 2009). Replacement of 9km of conductor and recloser replacement (Complete).
BDA2D06	BDA2D06 Bundarra Nth	265	186.6	30/06/2009	1208	1208	7.15	7.15	Pole failure on subtransmission, storms and vegetation.	Defect rectification and vegetation management.
BET1	BET1 Mt Ulandra Line 43	7	12.4	31/03/2009	1294	1414	6.00	8.00	Storms, wind and lightning.	A clearance problem has been rectified by putting delta constructions in the affected section. (Complete)
BGA2K13	BGA2K13 Horton Valley	172	132.0	31/12/2008	1968	1483	5.87	4.94	Pole failed, high winds and storms.	Pole replacement and defect rectification. Installation of 2 reclosers.(Completion September 2009)
BGA2K17	BGA2K17 Bingara Town/Ke era Rd	785	148.6	31/12/2008	1154	1006	5.43	6.11	Pole failed, high winds and storms.	Feeder due for inspection in Aug-09. Pole replacement and defect rectification. Installation of 3 reclosers. Replacement of 1 km of copper conductor. (Completion June 2011)
BJN8B4	BJN8B4 M31 Cryon	35	91.6	30/09/2006	944	1566	10.26	12.03	Storms and wind.	Ground line inspection. (Complete)

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					ANC	Current	ANC	Current		
BODB2	BODB2 Bodalla	66	38.6	31/03/2009	560	863	9.06	10.08	Fauna and storms / lightning.	Investigation report to be finalised.
BODC2	BODC2 Potato Pt	484	32.0	31/03/2009	955	780	8.04	6.02	High Winds and storms / lightning.	Performance to be monitored. Investigating need for CCT to counter migratory bats. Sensitive earth fault setting to be increased from 4 to 10 amps. (Complete)
BON3B1	BON3B1 Peacock/ Bottle Ck	184	54.6	30/09/2006	1584	1322	11.0 1	9.86	Storms/Lightning caused outages on subtransmission lines and upstream Zone Substations. Insulator failure in Casino Zone. Vegetation Clearing Incident.	Maintain Normal Inspection & Maintenance, Vegetation Maintenance Cycle - 3 yrs. Perform regular subtransmission line aerial patrol. Bonalbo Zone Substation - Establish Auto TX change-over. Installation of 2 reclosers & replacement of conductor at Jacksons flat. (Completion June 2010)
BON3B2	BON3B2 Old Bonalbo	337	103.4	30/09/2006	2004	1705	11.8 6	12.89	Storms/Lightning caused outages on subtransmission lines and upstream Zone Substations. Insulator failure in Casino Zone. Fauna.	Maintain Normal Inspection & Maintenance, Vegetation Maintenance Cycle - 3 yrs. Perform regular subtransmission line aerial patrol. Installation of 7 reclosers & replacement of 5.2kms of conductor at Duck Creek. (Completion June 2010)
BRE8B1	BRE8B1 M54 Walgett	33	161.4	31/12/2006	1866	2737	5.06	14.55	Storms.	Line maintenance. Installation of 3 reclosers. (Complete)

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					ANC	Current	ANC	Current		
BYA7881	BYA7881 Yarras	333	134.5	31/12/2008	1311	1341	5.95	5.98	Fauna, vegetation and Zone transformer failure.	Vegetation management. Installation of reclosers. (Complete)
BYA7882	BYA7882 Comboyne	430	157.9	30/09/2006	2149	1320	13.92	7.77	Fauna, equipment failure and wind.	Refurbishment & maintenance (Completion October 2009). Vegetation management (Complete). Replacement of 5.9km of conductor. Installation of 3 gas switches. Installation of 6 reclosers, 2 voltage regulators and a 33kV CB (Completion December 2009).
BZAR606320	BZAR606320 M63 Gunnedah Rd and Goran	37	69.17	30/06/2009	1509	1509	10.19	10.19	Crossarm failure on subtransmission. Pole failure and storms.	Defect rectification. Construction of an alternate feed to Breeza Z/S from Gunnedah via Curlewis. (Completion June 2011)
BZAR606321	BZA6321 M63 Walhallow	128	115.40	30/06/2009	1714	1714	9.09	9.09	Crossarm failure on subtransmission. Third party contact and storms.	Defect rectification. Construction of an alternate feed to Breeza Z/S from Gunnedah via Curlewis. (Completion June 2011)
CBY804	CBY804 Channel Nine Rd	56	65.3	31/03/2009	1098	1116	1.48	1.23	Pole failure in lightning storm.	Pole & line defect management. Vegetation management. Installation of a recloser. Upgrade Hydraulic reclosers. (Completion December 2009)
CLY1523	CLY1523 Bore Pumpers	52	89.4	31/12/2007	1346	881	4.08	3.87	Storms and lightning.	Pole & line defect management. Vegetation management. (Complete) Installation of a 33kV recloser. Protection review to be carried out. (Completion June 2010)

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					ANC	Current	ANC	Current		
CMK4005	CMK4005 West	136	199.9	31/12/2006	1189	1412	5.11	8.54	High winds and stock issues affecting the 66kV radial feeder.	Installation of 2 reclosers (Complete).
CNK3B2	CNK3B2 Jones Island	135	36.8	30/06/2007	1108	1038	8.41	9.44	Storms & fauna.	Maintain Normal Inspection & Maintenance, Vegetation Maintenance Cycle - 3 yrs. Perform regular subtransmission line aerial patrol. Installation of 7 reclosers & replacement of 5.2kms of conductor at Duck Creek. (Completion June 2010)
COW42	COW42 Lachlan	193	120.9	31/12/2007	428	230	10.90	3.41	Storms and lightning.	Line maintenance. Installation of 3 reclosers. (Complete)
CPD2C03	CPD2C03 Copeton Dam	37	31.0	31/03/2008	1255	250	4.19	1.43	66kV Equipment failure.	Aerial inspection of 66 kV subtransmission line. Installation of a recloser. (Complete)
CRO5512	CRO5512 Woodhouselee	496	184.0	31/03/2008	1374	818	3.64	3.09	Lightning storm.	Live line inspection of the subtransmission to be carried out 08/09 (Complete). Subtransmission line to be refurbished 09/10. Replacement of 30kms of conductor on the distribution feeder. (Completion December 2009)
DPR1097	DPR1097 Bringagee	29	67.5	31/12/2008	1046	520	5.07	2.45	Storms.	Pole & line defect management. Vegetation management.

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					ANC	Current	ANC	Current		
DUG3B7	DUG3B7 Salisbury	315	117.0	31/03/2008	1153	251	3.22	1.13	Vegetation and equipment failure.	Vegetation management (Completion January 2010). Installation of a recloser and gas switch (Completion July 2010).
DUN3B3	DUN3B3 Nimbin	584	76.9	31/12/2008	1051	1130	4.33	4.24	Storms / lightning and birds contacting mains.	Maintain Normal Inspection & Maintenance, Vegetation Maintenance Cycle - 3 yrs. Installation of a recloser and construction of a feeder tie to Nth Kyogle feeder. (Completion December 2009)
DUN3B4	DUN3B4 Mt Nardi	657	112.5	31/03/2009	1313	1448	4.28	4.96	Storms, high winds, lightning and vegetation.	Maintain Normal Inspection & Maintenance, Vegetation Maintenance Cycle - 3 yrs. Installation of a recloser and construction of a feeder tie to Nimbin Feeder. (Completion June 2011)
DUN3B5	DUN3B5 Rosebank	850	90.2	30/06/2008	1011	1388	5.46	9.22	Distribution and subtransmission faults due to severe storms and lightning.	Maintain Normal Inspection & Maintenance, Vegetation Maintenance Cycle - 2 yrs. Construction of a feeder tie to the Bangalow feeder and replacement of 4.6 kms of conductor. (Completion June 2010)
EDSC2	EDSC2 Kiah	428	113.0	31/12/2008	1805	1908	5.78	6.87	Conductor failed and storms.	Installation of 2 reclosers as part of the Eden South Zone substation reconstruction works (Completion June 2010).
ELI3B6	ELI3B6 Central Lismore	2366	17.6	31/12/2007	455	76	8.11	0.98	Severe storms/lightning, subtransmission faults and fauna.	Maintain Normal Inspection & Maintenance, Vegetation Maintenance Cycle - 1 yrs. Construction of a feeder tie to the North Lismore Feeder. Commissioning of the Lismore Uni Zone will split the feeder

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					ANC	Current	ANC	Current		
										(completed). (Completion June 2011)
GDH42	GDH42 M73 Pullaming	101	120.9	30/06/2008	1092	1010	5.00	7.34	Storms and fauna.	Vegetation management and targeted line maintenance. (Complete) Replacement of 2km of conductor. Installation of 6 reclosers. Construction of New Curlewis feeder. (Completion June 2010)
GDH6301	GDH6301 M62 Curlewis	330	114.4	31/03/2009	1840	1965	8.28	9.33	Storms / lightning caused conductor failure. Vegetation.	Installation of a recloser. (Completion June 2011)
GDH6801	GDH6801 M68 Quia	92	105.5	31/03/2009	1584	1584	5.89	5.89	Storms / lightning high winds.	Installation of a recloser. (Completion June 2011)
GDH6901	GDH6901 M69 Emerald Hill	108	101.0	31/03/2008	1268	1102	3.59	3.89	Third party contact & lightning.	Installation of 3 reclosers. (Completion June 2011)
GDL3B5	GDL3B5 Industrial	54	0.0	31/12/2008	1164	1045	1.69	1.06	Storms	Replacement of section of conductor. (Completion June 2010) Build interconnector 2010/11.

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					ANC	Current	ANC	Current		
HIL1114	HIL1114 Lower Lachlan	70	81.0	31/03/2009	1269	3021	3.97	5.80	Pole and conductor failure in lightning storms.	Installation of 2 reclosers. (Complete)
KYW2199	KYW2199 Kywong Sth	64	104.0	31/12/2008	1003	542	6.97	2.75	Conductor tie failure.	Pole & line defect management. Vegetation management. Refurbishment of subtransmission line. (Complete)
LKC7782	LKC7782 Bonny Hills	193	4.7	30/06/2008	964	3	9.05	0.03	Fauna, vegetation, Third party contact and equipment failure.	Vegetation management (Complete). Protection review to be carried out (Complete). Replacement of 4.7km of conductor & refurbishment of spurs along mainline (2.8 km completed).
MDI685	MDI685 Blandford	289	185.4	31/03/2009	1035	840	7.76	6.08	HV conductor failure and third party contact.	Installation of 3 reclosers. Replacement of feeder cables at Murrurundi ZS. Replacement of 5.9 km of conductor. (Completion June 2010)
MGE3B1	MGE3B1 Bonalbo Rd	57	40.7	30/09/2008	1944	1952	7.05	7.74	Subtransmission faults and distribution conductor failure in storm conditions. Birds contacting mains.	Maintain Normal Inspection & Maintenance, Vegetation Maintenance Cycle - 3 yrs. Perform regular subtransmission line aerial patrol. Refurbishment of Mallanganee Zone and installation of a recloser. (Completion June 2010)
MGE3B3	MGE3B3 Mallanganee/Casino Rd	215	102.4	30/09/2008	1007	333	7.92	5.46	Subtransmission faults and distribution conductor failure in storm conditions. Birds contacting mains.	Maintain Normal Inspection & Maintenance, Vegetation Maintenance Cycle - 3 yrs. Perform regular subtransmission line aerial patrol. Refurbishment of Mallanganee Zone Substation. Installation of a recloser at

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					ANC	Current	ANC	Current		
										Hogarth Range. (Complete)
MGI2U24	MGI2U24 Mungindi Town	366	6.1	31/12/2007	1157	1837	7.08	6.07	HV conductor, Storms and lightning	Installation of a recloser. (Complete)
MLA678	MLA678 West/Nth Rural	273	182.5	30/09/2008	1002	1038	6.62	4.52	Equipment failure, lightning	Carry out aerial pole inspections on subtransmission line (Complete). Installation of 5 reclosers & 7 gas switches. (Completion June 2010) Replacement of 7.1 km of conductor. (Completion June 2011)
MON1012	MON1012 Monteagle Rural	235	159.7	31/03/2009	956	1049	9.61	10.56	Storms and lightning. Pole failures.	Protection review to be conducted. Installation of 3 reclosers. Defect rectification of the subtransmission feeder. Zone substation refurbishment (Completion July 2010).
MRE2M37	MRE2M37 Gingham Rd	158	194.14	31/12/2008	843	606	8.15	5.58	Zone transformer failure and storms.	Installation of a recloser and a sectionaliser (Completion June 2010). Replacement of 1km of conductor (Completion June 2011).
MUL3B1	MUL3B1 Goonengerry	522	80.0	31/03/2008	1398	2828	3.81	7.98	Vegetation blown into conductors, high winds, delays due to flooding.	Maintain Normal Inspection & Maintenance, Vegetation Maintenance Cycle - 3 yrs. Installation of a recloser & replacement of conductor at Upper Wilson's Creek Rd and Goonengerry Rd. (Completion June 2010)
MUL3B2	MUL3B2 Main Arm	649	93.0	30/06/2008	1165	1480	7.15	5.20	Storms / severe weather conditions.	Maintain Normal Inspection & Maintenance, Vegetation Maintenance Cycle - 3 yrs.

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					ANC	Current	ANC	Current		
										Installation of reclosers & replacement of conductor at Main Arm & Middle Pocket. (Completion June 2010) 2011)
MWN3B1	MWN3B1 Uki	1206	157.9	31/12/2008	841	616	8.06	6.40	Storms / lightning and fauna contacting mains	Maintain Normal Inspection & Maintenance, Vegetation Maintenance Cycle - 3 yrs. Installation of a recloser and replacement of copper conductor at various locations. (Completion June 2011)
MWN3B5	MWN3B5 Tumbulgu m	893	89.3	31/12/2008	1312	803	4.52	3.20	Storms / lightning and fauna contacting mains.	Maintain Normal Inspection & Maintenance, Vegetation Maintenance Cycle - 3 yrs. Installation of a recloser and replacement of copper conductor at various locations. (Completion June 2011)
NBI3B1	NBI3B1 M14 Spring Ck	73	109.5	31/12/2008	1415	1533	3.52	5.36	Zone transformer failure.	Installation of 2 reclosers. Replacement of 6km of conductor. (Completion June 2011)
NCK3B5	NCK3B5 Bowraville Rural	599	194.7	31/12/2008	1224	2024	5.19	7.13	Equipment failure.	Installation of a recloser. Replacement of 5.6km of conductor. (Completion October 2009)
NER3B2	NER3B2 New Farm Rd	68	0.0	31/12/2008	1030	247	5.19	2.01	Subtransmission fault.	Thermovision survey of the feeder. Protection review. Vegetation and defect rectification. Refurbishment of the subtransmission feeder. Installation of 3 reclosers. (Complete)

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					ANC	Current	ANC	Current		
NEVE22	CB22 Gin Gin	96	151.4	31/12/2007	1148	474	6.01	2.64	Storms and lightning. Vegetation	Line Maintenance. (Complete)
NGN3B3	NGN3B3 Nana Glen	405	67.6	31/12/2008	1213	1457	7.05	8.71	Vegetation and equipment failure.	Vegetation management. Installation of a recloser. Protection review to be carried out. Aerial inspection to be carried out. (Completion June 2010)
NHD3B4	NHD3B4 Valla	956	78.6	30/06/2008	1037	748	2.82	5.17	Vegetation, overload and vegetation.	Vegetation management. Protection review to be carried out. (Complete)
NY70	NYR70 Kangaroo Ck	48	34.2	31/12/2008	1821	3154	10.05	11.77	Storms / lightning and fauna contacting mains.	Maintain Normal Inspection & Maintenance, Vegetation Maintenance Cycle - 3 yrs. Refurbishment of Nymboida Zone & installation of a recloser at Kangaroo Creek Road. (Complete)
OKY3B2	OKY3B2 Jeogla	144	103.9	31/03/2007	1352	1163	11.76	5.45	Storms and lightning.	Vegetation management and line maintenance. Installation of 2 sectionalisers (Completion June 2010). Replacement of 17.5 km of conductor. Protection upgrade. (Completion June 2011)
OWES941	CB941 Cargo Rd	512	142.5	31/12/2008	1004	1106	7.05	6.64	Storm, third party contact, vegetation and equipment failure.	Protection review to be conducted. Installation of a recloser (Completion July 2010).

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Appendix B - Table 4.12: Rural Short Feeders Which Exceeded the Individual Feeder Standards During the Previous 12 Month Period

SAIDI Standard: 1000; SAIFI Standard: 8

Feeder Name	Location	Customer Numbers	Length	Date of First Non-Compliance	SAIDI		SAIFI		Description of Non-Compliance and Reason	Remedial Actions Proposed or Taken Including Timetable
					ANC	Current	ANC	Current		
PCP14212	PCP14212 2 Coomba Pk	607	67.3	30/09/2008	1277	1342	5.14	5.02	Equipment failure, fauna and vegetation.	Vegetation management (Complete). 11kV exit cable from new switchboard for Coomba Park 11kV feeder . Installation of a recloser. (Completion July 2010)
QUI3B2	QUI3B2 Mogilla	163	91.9	30/09/2008	1323	1431	8.29	8.10	Equipment failure and wind.	Defect rectification. Vegetation maintenance. (Complete) Installation of a recloser on the Tantawangalo spur.
RCF10	RCF10 Brooms Head	316	15.6	30/09/2006	1392	47	11.01	1.01	Pole failure.	Maintain Normal Inspection & Maintenance, Vegetation Maintenance Cycle - 2 yrs. Replacement of conductor on the 33KV subtransmission line. (Completion March 2011)
RGL3B11	RGL3B11 Raleigh/Repton	560	26.8	30/06/2009	742	742	11.84	11.84	Storms & wind. Subtransmission line outages while network in abnormal configuration.	Construction of New Boambee 132/66/11kV Zone Substation. (Completion November 2009)
RGL3B5	RGL3B5 Urunga No.1	1207	14.9	30/06/2009	610	610	9.16	9.16	Storms & fauna. Subtransmission line outages while network in abnormal configuration.	Construction of New Boambee 132/66/11kV Zone Substation. (Completion November 2009)
RGL3B7	RGL3B7 Bellingen Rural	1266	157.6	31/03/2008	751	1601	9.90	10.13	Equipment failure, storms, wind, vegetation and fauna.	Vegetation management (Complete). Replacement of conductor (Complete). Installation of a recloser (Complete). New Zone Sub planned (Completion November 2009).

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					ANC	Current	ANC	Current		
SGN3B6	SGN3B6 Rushforth Rd	1563	78.9	30/09/2008	559	128	8.18	3.31	Storms / lightning and birds contacting mains.	Maintain Normal Inspection & Maintenance, Vegetation Maintenance Cycle - 3 yrs. Replacement of 5.3kms of conductor at Coutts Crossing. (Completion December 2009)
THA2224	THA2224 Bromley Rd	85	81.2	30/06/2007	1281	288	4.33	1.89	Recloser and pole failure, storms.	Pole & line defect management. Vegetation management. Installation of 3 reclosers. Replacement of conductor. (Complete)
THR	THR Sewerage Treatment	1	0.2	30/09/2008	1488	1757	1.00	3.00	Underground cable fault - heavy snow delayed restoration.	Performance to be monitored.
TPT3B3	TPT3B3 Rollands Plains	330	112.9	30/06/2008	1651	765	8.62	4.15	Bush fire, vegetation and high winds.	Vegetation management & aerial patrol. Protection review to be carried out. Replacement of 9.8km of conductor. Refurbishment of spurs along mainline. Installation of 6 reclosers. (Completion June 2010)
UBN3B2	UBN3B2 Urbenville /Sth Boomi	211	66.4	30/09/2006	1739	1268	12.20	11.91	Storms/lightning causing outages on subtransmission lines and upstream Zone Substations. Insulator failure in Casino Zone.	Maintain Normal Inspection & Maintenance, Vegetation Maintenance Cycle - 3 yrs. Perform regular subtransmission line aerial patrol. Refurbishment of Urbenville Zone Substation. Installation of 3 reclosers & ganged fuses. (Completion June 2010)

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					ANC	Current	ANC	Current		
UBN3B3	UBN3B3 Woodenbong/Boomi Ck	309	57.0	30/09/2006	1539	1228	14.24	11.01	Storms/lightning causing outages on subtransmission lines and upstream Zone Substations. Insulator failure in Casino Zone.	Maintain Normal Inspection & Maintenance, Vegetation Maintenance Cycle - 3 yrs. Perform regular subtransmission line aerial patrol. Refurbishment of Urbenville Zone Substation. Installation of recloser & ganged fuses (Complete).
WBN3B2	WBN3B2 Swan Bay	396	144.6	30/09/2008	1015	1261	8.53	7.36	Storms/lightning and polymer insulator failure on radial 66KV line. Zone transformer failure.	Maintain Normal Inspection & Maintenance, Vegetation Maintenance Cycle - 3 yrs. Replacement of two 66KV circuit breakers and installation of two 8/10 MVA 66/11KV transformers. Investigate installation of overhead earthwire on Woodburn 66KV line. Installation of a recloser, reconductoring and Tie-Line.(Completion December 2010)
WBN3B3	WBN3B3 Coraki	821	98.7	30/09/2008	977	752	9.37	5.30	Storms/lightning and polymer insulator failure on radial 66KV line. Zone transformer failure.	Maintain Normal Inspection & Maintenance, Vegetation Maintenance Cycle - 3 yrs. Zone Substation project as per WBN3B2. Installation of a Recloser & reconductoring. (Completion June 2010)
WBN3B4	WBN3B4 Woodburn	342	37.6	31/03/2009	1137	1076	5.72	4.72	Storms/lightning and polymer insulator failure on radial 66KV line. Zone transformer failure. Bush fire and fauna.	Maintain Normal Inspection & Maintenance, Vegetation Maintenance Cycle - 3 yrs. Zone Substation project as per WBN3B2. Installation of a recloser & 1.9km of reconductoring. (Completion June 2010)

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					ANC	Current	ANC	Current		
WBN3B5	WBN3B5 Evans Head Sth	949	21.2	29/09/2008	859	997	8.02	6.30	Storms/lightning and polymer insulator failure on radial 66KV line. Zone transformer failure.	Maintain Normal Inspection & Maintenance, Vegetation Maintenance Cycle - 2 yrs. Zone Substation project as per WBN3B2. Removal of a river crossing and replacement of conductor at various locations at Evans Head. (Completion June 2010)
WBN3B6	WBN3B6 Evans Head Nth	877	28.0	30/09/2008	962	613	8.06	4.03	Storms/lightning and polymer insulator failure on radial 66KV line. Zone transformer failure.	Maintain Normal Inspection & Maintenance, Vegetation Maintenance Cycle - 2 yrs. Zone Substation project as per WBN3B2.
WBN3B7	WBN3B7 Broadwater	308	30.5	30/09/2008	933	913	8.40	6.43	Storms/lightning and polymer insulator failure on radial 66KV line. Zone transformer failure.	Maintain Normal Inspection & Maintenance, Vegetation Maintenance Cycle - 3 yrs. Zone Substation project as per WBN3B2. Replacement of conductor at Broadwater (Completion April 2010).
WGA3B3	WGA3B3 Wooli	1194	155.9	30/06/2009	1010	1010	6.47	6.47	Storms, wind, fauna and broken insulators.	Investigation report to be prepared.
WGT342	WGT342 M34 Lightning Ridge	35	127.2	31/12/2007	1098	1089	4.66	8.37	Equipment failure. Fauna.	Installation of a recloser. (Complete)
WLGR4236	WLGR42 36 Limestone	63	98.00	30/09/2008	828	447	8.06	2.78	Storms.	Performance to be monitored.

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					ANC	Current	ANC	Current		
WNA2V05	WNA2V05 Bullarah	51	119.7	31/03/2007	1647	514	10.39	4.16	Lightning, storms and fauna.	Installation of a recloser. Replacement of 10.4 km of conductor (Completion June 2010). Construction of Curragundi Tie feeder (Completion June 2011).
WTR2Z07	WTR2Z07 Moomin Ck	33	72.8	31/03/2007	1231	942	10.33	6.55	Lightning and storms.	Performance to be monitored.
WWA72	WWA72 M9 Merah Nth/Dore en Ln	123	142.7	31/03/2007	1335	724	6.46	4.60	Storms.	Installation of 2 reclosers. (Completion June 2010)
WWAM7	WWAM7 Yarrie Lake	111	166.9	30/06/2009	1310	1310	6.38	6.38	Fauna and storms.	Installation of 2 sectionalisers and replacement of conductor. (Completion October 2009)
YEN3B2	YEN3B2 East Yenda	97	13.1	31/03/2009	1022	872	3.08	2.47	HV conductor failure in lightning storm and fauna.	Construction of new Subtransmission line to Yenda Z/S. (Completion November 2009)
YEN3B7	YEN3B7 Mt Bingar	462	97.0	31/03/2009	1008	919	3.36	3.24	HV conductor failure in lightning storm and fauna.	Construction of new Subtransmission line to Yenda Z/S. (Completion November 2009)

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					ANC	Current	ANC	Current		
YLI2H13	YLI2H13 Gunyerwarildi	56	111.9	31/12/2008	1103	644	4.64	3.13	Fauna and storms.	Installation of 2 reclosers. (Complete)

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Appendix B - Table 4.13: Rural Long Feeders Which Exceeded the Individual Feeder Standards During the Previous 12 Month Period

SAIDI Standard: 1400; SAIFI Standard: 10

Feeder Name	Location	Customer Numbers	Length	Date of First Non-Compliance	SAIDI		SAIFI		Description of Non-Compliance and Reason	Remedial Actions Proposed or Taken Including Timetable
					ANC	Current	ANC	Current		
AHY2T04	AHY2T04 Northern	220	425.7	30/06/2009	1462	1462	6.46	6.46	Storms & wind. Third Party contact - aircraft.	Defect rectification in progress.
ANN8B1	ANN8B1 No.1 Fdr Line 72	403	465.4	31/12/2007	1805	988	5.01	3.43	Long SWER lines subject to storms and lightning.	Installation of 10 reclosers (Complete).
BJN8B2	BJN8B2 M9 Carbeen/ Merah Nth	114	274.4	30/06/2007	1474	1817	6.88	11.67	Storms and lightning. Insulator failure	Defect rectification and protection review (Complete). Installation of 4 reclosers (Completion June 2011).
BJN8B3	BJN8B3 M37 Yarran	264	367.6	31/12/2007	1488	2334	5.82	13.53	Wind, storms and lightning.	Defect rectification. Vegetation management. (Complete) Installation of 4 reclosers (Completion June 2011).
BJN8B5	BJN8B5 M30 Collarene bri	83	241.4	30/09/2006	1404	3813	10.18	17.35	Storms and lightning.	Ground line inspection. Installation of 2 reclosers. (Complete)
BKE4B1	BKE4B1 M81 Enngonia	115	553.7	30/06/2007	1083	2382	18.18	12.50	Storms, pole failure and fauna (birds).	Installation of 11 sectionalisers, 5 reclosers, replacing 24 HV fuse links with solid links. (Completion November 2009)
BKE4B2	BKE4B2 M82 Wanaarin g	252	1828.6	31/03/2007	1700	2962	5.72	13.77	Storms and lightning.	Defect rectification. Installation of 2 reclosers & replacement of 130 poles with steel poles. (Complete)

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Feeder Name	Location	Customer Numbers	Length	Date of First Non-Compliance	SAIDI		SAIFI		Description of Non-Compliance and Reason	Remedial Actions Proposed or Taken Including Timetable
					ANC	Current	ANC	Current		
BKE4B3	BKE4B3 M83 Louth	98	598.0	30/06/2009	1188	1188	10.57	10.57	Storms and crossarm failure.	Investigation report to be developed.
BRE8B2	BRE8B2 M51 Goodooga	125	693.1	31/03/2007	1708	1962	6.68	5.56	Storms and lightning.	Defect rectification. Installation of 3 reclosers. (Complete)
BRE8B7	BRE8B7 M53 Collerina	40	299.0	31/03/2007	2019	2131	9.65	4.80	Storms and lightning.	Pole replacement and defect rectification (Complete).
CDN62	CB62 Tallebung Original Fdr 065	212	858.9	31/03/2007	1668	745	9.97	4.00	Storms, voltage regulators failed during storms.	Installation of a recloser and replacement of 2 regulators (Complete).
CR05502	CR05502 Narrawa Bevendale	459	229.6	30/06/2008	1444	776	4.67	2.73	Lightning storm.	Live line inspection of the subtransmission to be carried out 08/09 (Complete). Subtransmission line to be refurbished 09/10 (Completion December 2009).
CR05522	CR05502 2 Tuena Redground	638	428.0	31/03/2008	1594	781	5.63	2.68	Lightning storm.	Live line inspection of the subtransmission (Complete). Subtransmission line to be refurbished (Completion December 2009).
CS03B2	CS03B2 Kyogle Rd	948	268.0	30/06/2009	1965	1965	5.95	5.95	Storms and high winds	Maintain Normal Inspection & Maintenance, Vegetation Maintenance Cycle - 3 yrs. Installation of a recloser and replacement of conductor at various locations. (Completion June 2010)

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Feeder Name	Location	Customer Numbers	Length	Date of First Non-Compliance	SAIDI		SAIFI		Description of Non-Compliance and Reason	Remedial Actions Proposed or Taken Including Timetable
					ANC	Current	ANC	Current		
GID42	GID42 Rural Nth	172	204.6	30/06/2008	1555	488	7.16	5.85	Storms and lightning.	Line maintenance. (Complete)
HIL1111	HIL1111 Merriwagga	197	553.6	30/06/2009	1987	1987	5.28	5.28	Storms	Investigation report to be developed.
IVA1326	IVA1326 North West SWER	40	230.5	31/12/2007	1439	531	3.02	2.83	HV conductor, storms and lightning.	Pole & line defect management. Vegetation management. Protection review to be carried out. (Complete)
IVL2R22	IVL2R22 Kings Plains	424	275.9	31/03/2009	1008	967	10.6 1	9.72	Storms / lightning caused conductor failure. Fauna.	Pole inspection. Replacement of 1.2km of conductor. Installation of a recloser. (Completion June 2010)
IVL2R23	IVL2R23 Bukulla/Wallangra	632	592.0	31/03/2008	1169	568	10.7 1	4.99	Storms/lightning, vegetation.	Vegetation management, targeted line maintenance and pole replacements. Installation of 6 reclosers, regulators at Wallangra Zone Substation and replacement of 4.5km of conductor. (Complete)
LGR342	LGR342 M34 Goodooga	283	390.0	31/12/2007	1818	1322	3.66	3.48	Storms and lightning.	Line maintenance. Installation of a recloser. (Complete)
MGA0682	MGA0682 Willawarren	631	315.9	30/09/2008	1402	955	7.62	5.55	Equipment failure & fauna.	Vegetation management. Replacement of 14.7km of conductor (Completion June 2010). Installation of 2 reclosers (Complete).
MGE3B2	MGE3B2 Tenterfiel	460	238.5	30/09/2006	1205	869	11.0 0	8.90	Subtransmission faults and distribution	Maintain Normal Inspection & Maintenance, Vegetation Maintenance Cycle - 3 yrs. Perform

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					ANC	Current	ANC	Current		
	d Rd/Deep Ck								conductor failure in storm conditions.	regular subtransmission line aerial patrol. Refurbishment of Mallanganee Zone Substation. Installation of 3 reclosers & replacement of 1.3km of conductor. Construction of a 4 km feeder tie at Paddys Flat Rd. (Completion June 2010)
MGI2U23	MGI2U23 Boomi Rd	191	338.8	30/09/2008	1611	2130	8.64	10.01	Storms, fauna & equipment failure.	Carry out aerial pole inspections on subtransmission line. Refurbish DC supply at Mungindi Z/S. Replacement of 12.6km of conductor. Replacement of 3 sectionalisers. (Completion October 2009)
MGI2U30	MGI2U30 Curragundi	155	413.1	30/09/2006	1545	1981	6.96	8.08	Conductor, Insulator and crossarm failure	Vegetation management, targeted line maintenance and pole replacements. Installation of 6 reclosers. (Complete)
MRE2M35	MRE2M35 Eastern Rural	303	273.28	30/09/2007	1480	1005	10.37	8.55	High winds, faulty ABS, conductor and transformer failures.	Installation of 4 reclosers & 11 gas switches. (Completion June 2010)
MRE2M38	MRE2M38 Tareelaro	191	312.9	31/12/2007	1443	554	8.24	4.66	Storms and fauna.	Pole inspections. Installation of 3 reclosers & 10 gas switches. (Completion December 2009)
NEVE42	CB42 Warren Rural Nth	402	795.9	31/12/2007	1489	1391	8.27	7.04	Storms, lightning and wind.	Performance to be monitored.
OBR52	OBR52 Black Springs	382	308.3	31/03/2009	1512	1522	6.57	6.00	Vegetation blown into conductors during storm.	Investigation report to be finalised.

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RAG3B6	RAG3B6 Limekilns Rd	946	267.7	31/03/2007	608	50	12.4 2	0.48	Severe storms.	Line patrol and defect rectification. Circuit breaker maintenance to be carried out (Complete). Installation of a recloser (Complete).
TXS3X06	TXS3X06 Yelarbon	325	487.60	30/06/2008	1527	307	11.0 4	3.85	Storms and equipment failure.	Pole inspection (Completion October 2009). Installation of 7 reclosers and 3 gas switches (Completion June 2010). Replacement of 14.4km of conductor (Completion June 2011).
WGT322	WGT322 M32 Come-By-Chance	125	315.2	31/03/2007	2394	5181	6.46	18.35	Severe lightning storms.	Defect rectification. Installation of 4 sectionalisers. (Complete)
WGT332	WGT332 M33 Collarene bri	68	307.8	30/09/2006	1575	2322	7.77	10.78	Severe lightning storms.	Line maintenance. Installation of a recloser. (Complete)
WGT352	WGT352 M35 Cumborah	100	353.1	31/03/2009	1600	2470	8.96	14.46	Lightning and storms caused equipment failure. Fauna.	Investigation report to be updated.
WGT362	WGT362 M36 Carinda	146	346.0	31/03/2008	1775	3597	6.95	8.77	Storms and equipment failure.	Performance to be monitored. Protection review to be carried out. Installation of a recloser and 3 sectionalisers. (Completion August 2009)
WGT392	WGT392 M38 and	93	274.9	31/03/2007	2568	6268	8.69	11.71	Storms and lightning.	Defect rectification. Installation of 4 sectionalisers. (Complete)

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					ANC	Current	ANC	Current		
	M39 Pilliga									
WGT542	WGT542 M54 Brewarrina	41	250.1	30/09/2008	2042	6053	10.95	30.98	Storms and lightning.	General maintenance and defect rectification.(5 defects remain) Protection review undertaken. Installation of 3 reclosers and 2 sectionalisers. Use of steel poles as replacements to combat termite problems. (Completion September 2009)
WJEM32	CB32 West Plain	160	319.9	30/06/2009	2345	2345	4.76	4.76	Transformer failure due to storms and lightning.	Performance to be monitored.
WTR2Z08	WTR2Z08 Poison Gate	86	215.2	30/09/2006	1610	885	9.12	7.63	Lightning, storms and high winds.	Installation of a recloser and a sectionaliser (Completion June 2010). Replacement of 6.3km of conductor (Completion June 2011).